

STUDY MATERIAL

COMPUTER FUNDAMENTALS

- **17 MARKS IN Artificial Intelligence [ARTI] ----- CLASS XI**
- **17 MARKS IN DATA SCIENCE [DTSC] ----- CLASS XI**
- **MAY ALSO BE USED BY STUDENTS OF COMA AND COMS**

Syllabus for Section 1 : Computer Fundamentals [17 Marks] –ARTI & DTSC

- **History of computer ,Basic Computer hardware, input and output devices, Basic Computer architecture , Memory and CPU , Networking of Machines (overview of LAN , MAN , WAN , Internet , Wifi etc) , types of computer (workstation, desktop, smartphone, embedded system etc.), Overview of software (system software and Application software with examples), Definition of Operating system and functions.**
- **Bit,Byte and Word, Number system (Base, Binary , Decimal , Octal , Hexadecimal), Conversion of Number systems , Boolean Logic (Boolean gates) , Boolean operators, ASCII code , Concept of Algorithm and Flowchart.**
- **Basics of Computer Programming (three levels: high level language , assembly language , machine language , definition and block diagrams) , overview of compiler and interpreter (definition and mention names of major compiled and interpreted languages) , overview of procedural and object-oriented programming(key features and just the basic differences , mention names of some popular procedural and object-oriented languages)**

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History of Computers

Before computers were developed people used sticks, stones, and bones as counting tools. As technology advanced and the human mind improved with time more computing devices were developed like Abacus, Napier's Bones, etc. These devices were used as computers for performing mathematical computations but not very complex ones.

The concept of computation dates back to ancient civilizations. The abacus, developed around 2000 BCE, was one of the earliest tools used for arithmetic calculations. Other devices, such as the Antikythera mechanism from ancient Greece, demonstrated a remarkable understanding of astronomical calculations. Some of the popular computing devices are described below, starting from the oldest to the latest or most advanced technology developed:

Abacus

Around 4000 years ago, the Chinese invented the Abacus, and it is believed to be the first computer. The history of computers begins with the birth of the abacus.

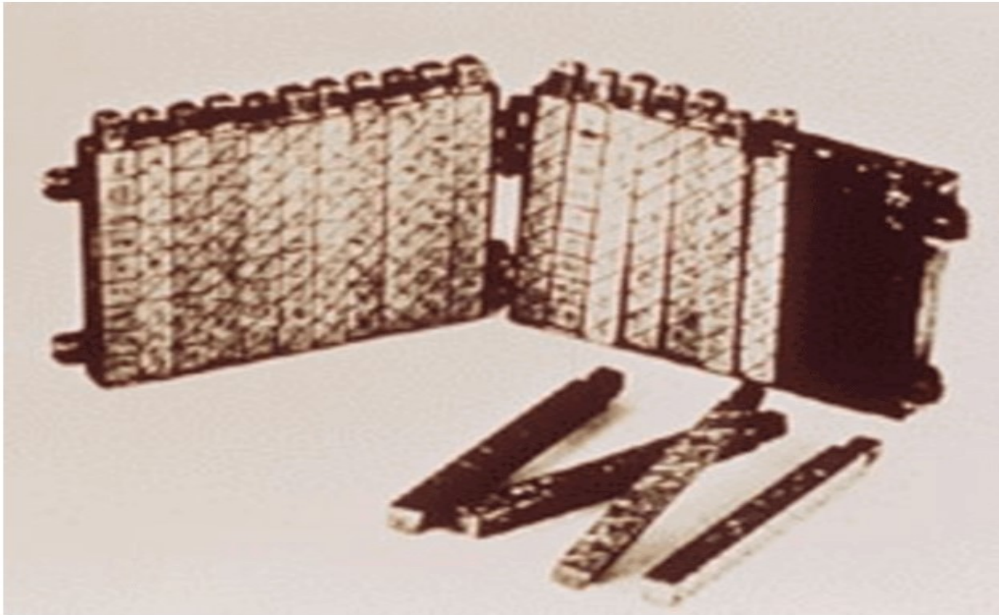
Structure: Abacus is basically a wooden rack that has metal rods with beads mounted on them.

Working of abacus: In the abacus, the beads were moved by the abacus operator according to some rules to perform arithmetic calculations. The beads represented the columns. The right-most column represented the unit, the next for ten and so on. Numbers were represented by moving beads at appropriate column. Abacus could perform simple addition and subtraction.



Napier's Bones

Napier's Bones was a manually operated calculating device and as the name indicates, it was invented by John Napier. In this device, he used 9 different ivory strips (bones) marked with numbers to multiply and divide for calculation. It was also the first machine to use the decimal point system for calculation.



Pascaline

It is also called an Arithmetic Machine or Adding Machine. A French mathematician-philosopher Blaise Pascal invented this between 1642 and 1644. It was the first mechanical and automatic calculator. It was invented by Pascal to help his father, a tax accountant, in his work or calculation. It could perform addition and subtraction in quick time. It was basically a wooden box with a series of gears and wheels. It is worked by a rotating wheel like when a wheel is rotated one revolution, it rotates the neighbouring wheel and a series of windows is given on the top of the wheels to read the totals.



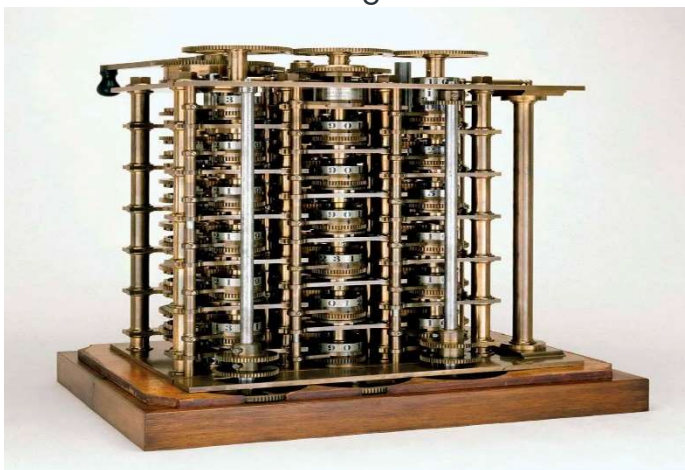
Stepped Reckoner or Leibniz wheel

A German mathematician-philosopher Gottfried Wilhelm Leibniz in 1673 developed this device by improving Pascal's invention to develop this machine. It was basically a digital mechanical calculator, and it was called the stepped reckoner as it was made of fluted drums instead of gears (used in the previous model of Pascaline).



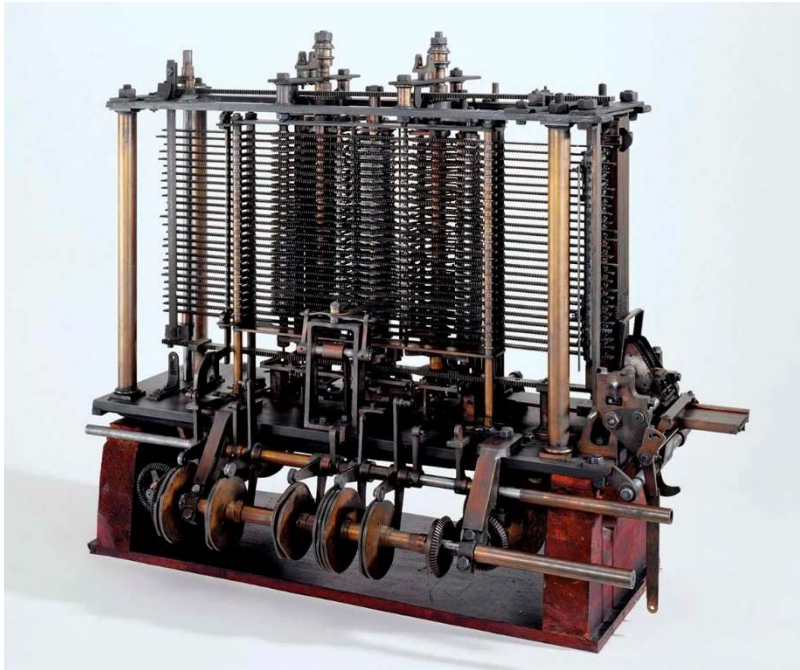
Difference Engine

Charles Babbage who is also known as the “Father of Modern Computer” designed the Difference Engine in the early 1820s. Difference Engine was a mechanical computer which is capable of performing simple calculations. It works with help of steam as it was a steam-driven calculating machine, and it was designed to solve tables of numbers like logarithm tables.



Analytical Engine

Again in 1830 Charles Babbage developed another calculating machine which was Analytical Engine. Analytical Engine was a mechanical computer that used punch cards as input. It was capable of performing or solving any mathematical problem and storing information as a permanent memory (storage).



Tabulating Machine

Herman Hollerith, an American statistician invented this machine in the year 1890. Tabulating Machine was a mechanical tabulator that was based on punch cards. It was capable of tabulating statistics and record or sort data or information. This machine was used by U.S. Census in the year 1890. Hollerith's Tabulating Machine Company was started by Hollerith and this company later became International Business Machine (IBM) in the year 1924.



Differential Analyzer

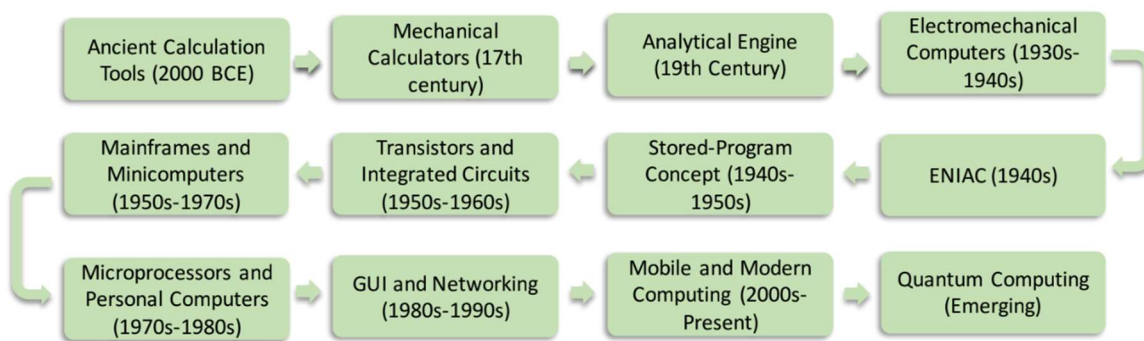
Differential Analyzer was the first electronic computer introduced in the year 1930 in the United States. It was basically an analog device that was invented by Vannevar Bush. This machine consists of vacuum tubes to switch electrical signals to perform calculations. It was capable of doing 25 calculations in a few minutes.

Mark I

In the year 1937, major changes began in the history of computers when Howard Aiken planned to develop a machine that could perform large calculations or calculations involving large numbers. In the year 1944, Mark I computer was built as a partnership between IBM and Harvard. It was also the first programmable digital computer marking a new era in the computer world.



History of Computers



Generations of Computers

First Generation Computers [1940–1956] : Vacuum Tubes

These machines were slow, huge, and expensive. In this generation of computers, vacuum tubes were used as the basic components of CPU and memory. Also, they were mainly dependent on the batch operating systems and punch cards. Magnetic tape and paper tape were used as output and input devices. For example ENIAC–Electronic Numerical Integrator and Calculator , UNIVAC-1– Universal Automatic Computer , EDVAC– Electronic Discrete Variable Automatic Computer, etc.

Second Generation Computers [1957–1963] : Transistors

It was the time of the transistor computers. In the second generation of computers, transistors (which were cheap in cost) were used. Transistors are also compact and consume less power. Transistor computers are faster than first-generation computers. For primary memory, magnetic cores were used, and for secondary memory magnetic disc and tapes for storage purposes. In second-generation computers, COBOL and FORTRAN were used as Assembly language and programming languages, and Batch processing and multiprogramming operating systems were used in these computers.

For example IBM 1620, IBM 7094, CDC 1604, CDC 3600, etc.

Third Generation Computers [1964 – Early 1970s] : Integrated Circuits

In the third generation of computers, integrated circuits (ICs) were used instead of transistors(in the second generation). A single IC consists of many transistors which increased the power of a computer and also reduced the cost. The third generation computers were more reliable, efficient, and smaller in size. It used remote processing, time-sharing, and multiprogramming as operating systems. FORTRAN-II TO IV, COBOL, and PASCAL PL/I were used which are high-level programming languages.

For example IBM-360 series, Honeywell-6000 series, IBM-370/168, etc.

Fourth Generation Computers [Early 1970s – Till date] : Microprocessors

The period of 1971–1980 was mainly the time of fourth generation computers. It used VLSI(Very Large Scale Integrated) circuits. VLSI is a chip containing millions of transistors and other circuit elements and because of these chips, the computers of this generation are more compact, powerful, fast, and affordable(low in cost). Real-time, time-sharing and distributed operating system are used by these computers. C and C++ are used as the programming languages in this generation of computers.

For example STAR 1000, PDP 11, CRAY-1, CRAY-X-MP, etc.

Fifth Generation Computers [Present and Beyond] : Artificial Intelligence

The ULSI (Ultra Large Scale Integration) technology is used in fifth-generation computers instead of the VLSI technology of fourth-generation computers. Use of fifth generation computers are currently in research level. Microprocessor chips

with ten million electronic components are used in these computers. Parallel processing hardware and AI (Artificial Intelligence) software are also used in fifth-generation computers. The scientists are trying to use AI and KIPS to perform any complicated problem. The programming languages like C, C++, Java, .Net, etc. are used.

For example Desktop, Laptop, NoteBook, UltraBook, etc.

Types of Computers

Classification of generations of computers

Generation	Timeline	Evolving hardware	Main memory	Programming language	Input/output devices	Examples
First	1940s-1950s	Vacuum tube based	Magnetic drums and tapes	Machine language	Punched cards and paper tape.	ENIAC, UNIVAC1, IBM 650, etc
Second	1950s-1960s	Transistor based	Magnetic core and magnetic tape / disk	Assembly language	Punched cards and magnetic tape	IBM 1401, IBM 7090 and 7094, UNIVAC 1107, etc.
Third	1960s-1970s	Integrated circuit based	Magnetic core and magnetic tape / disk	High level language (FORTRAN, BASIC, Pascal, COBOL, C)	Magnetic tape, keyboard, monitor, printer, etc.	IBM 360, IBM 370, PDP-11, UNIVAC 1108, etc.
Fourth	1970s-present	Microprocessor based	RAM, ROM	High level language (Python, C++, Java,.)	Keyboard, pointing devices, optical scanning, monitor, printer, etc.	IBM PC, STAR 1000, APPLE II, Apple Macintosh, etc.
Fifth	The present and the future	Artificial intelligence based	RAM, ROM	Understand natural language (human language)	Keyboard, monitor, mouse, trackpad touchscreen, pen, speech input, light scanner, printer, etc.	Desktops, laptops, tablets, smartphones, etc.

based on Data Type

Based on the data type handling, computers can be categorized as Digital, Analog, and Hybrid.

Digital

Personal computers are an example of a digital computer. These computers accept input in the form of 0s and 1s. The computer processes binary input and provides the output. These computers perform all the logical & arithmetical operations. Any input given in any language is first converted into binary language and then the computer processes the information. Examples – laptops, PCs, mobile phones, desktops, etc.

Analog

These computers process analog data. Analog data keep varying. Hence, it does not have any discrete value. They read the continuous change in the input, process it, and then provide the output. Analog computers perform with equal diligence and accuracy. They are however slower than digital computers. They are also slightly less precise.

Analog computers are for a Speedometer, thermometer, frequency, and signal of voltage, measuring the resistance of a capacitor.

Hybrid

Hybrid computers are a mix of both analog and digital computers. These computers perform a high level of calculations. Hybrid computers are quick and efficient. They take input in analog form, convert it into digital form, and then process it to produce an output. Scientists are also using hybrid computers for complex calculations. For example, in hospitals to measure the heartbeat of the patients, and at research institutes to measure earthquakes and other natural calamities.

Based on the Purpose

Microcomputer

Microcomputers are nothing but personal computers. These are single-chip systems. These are useful for personal use and can perform all the basic functions of the computer. Microcomputers require very little space and are comparatively inexpensive. Such computers have the most minimalistic requirement in terms of I/O devices. And have all the circuitry mounted on a single PCB. For example tablets, I pads, smartwatches, laptops, desktops

Minicomputer

Standing in between a microcomputer and a mainframe computer is the minicomputer. These computers are useful if people around 5 to 300. Those who want to operate the system at the same time. You can see such computers at the billing counters of malls or large institutions.

Mainframe

Mainframe computers are useful when a large number of people are involved. Like in the health care or retail sector who want to access data simultaneously. These computers process large amounts of data.

In addition, mainframe computers have evolved a lot over the years in terms of speed, size, and efficiency. These computers are just below the supercomputers. And sometimes are even more useful than a supercomputer. Examples – IBM z Series, System z9, etc.

Supercomputers

The biggest and fastest computers are supercomputers. Such computers can process trillions of functions within a few seconds. We generally use MPIS(Million Instructions Per Second) to measure their performance. These computers are specifically designed for scientific applications such as –

1. Encryption decryption of passwords
2. Weather forecasting
3. Testing of nuclear weapons
4. Scientific research of earth and other planetary systems, etc.

Examples – [PARAM supercomputer series](#), Gravity Pipe for astrophysics, Deep Crack for deciphering codes, etc.

Based on the Functionality

Workstations

Workstation computers are for single usage and professional purposes. These are like our basic laptops and desktops but with added superior features. For example, double-processor motherboard, added graphic card, ECC RAM, etc. The workstations are more powerful as compared to generic PCs. These can handle heavy-duty functions. Like animation, CAD, audio & video editing, professional gaming, etc.

Examples: Apple PowerBook G4, SPARC CPU, MIPS CPU, etc.

Servers

These are hardware components or software programs that are built to assist other computers termed as clients. Together this architecture is called the client-server model. The client sends a request to the server and the server responds in return with a result or a solution. This proves that these server computers are more powerful than standard computers. The main purpose of these computers is to share data and resources with other computers. Different types of servers are useful for different needs and applications. For example, cloud server, application server, database server, file server, etc. Each of these servers has a different purpose for different client needs.

Embedded

These computers are mainly microcontroller-based systems. Used for processing specific tasks. Embedded computers have a combination of software and hardware components. But, are usually a part of a larger system. Each of its components is designed from scratch to serve a specific purpose or complete a specific task. Another characteristic that distinguishes them from a standard PC is that all of its components are integrated into a single PCB or motherboard. They are most helpful for industrial use. this is because of their ruggedness. Examples: GPS systems, centralized heating systems, fitness trackers, digital watches, electronic calculators, etc.

Information appliances

Mostly portable devices designed for specific functions come under this category. They can perform very restricted tasks for which they are built like text editors, music players, photography, videography, etc. The most common example is mobile phones. Many wearable devices are also available in the market.

This completes our article on the different types of computers. To sum up, it can be said that there is everything for everyone.

Edge Devices

An edge device is any piece of hardware that controls data flow at the boundary between two networks. Edge devices fulfill a variety of roles, depending on what type of device they are, but they essentially serve as network entry -- or exit -- points.

Traditional edge devices include edge routers, routing switches, firewalls, multiplexers, and other wide area network (WAN) devices. Intelligent edge devices have built-in processors with onboard analytics or artificial intelligence capabilities. Such devices might include sensors, actuators, and IoT gateways.

Input and Output Devices of Computer with Examples

Nowadays computer directly or indirectly has become a part of our life from banking accounts to business deals. Do you know, How a computer works? Both input and output devices are important for a computer to work well and be easy to use. With the help of the input devices, you can send instructions to the system of the computer to execute an action based on your needs. With the help of the output devices, you can get the result or outcome of your inserting instructions on the screen or other output platforms. This article will provide you with information about the different input and output devices in a Computer system.

Examples of Input Devices of Computer	
Keyboard	Mouse
Joysticks	Wii Remote
Light Pen	Game pad
Microphone	Webcam
Scanner	Digital camera
Barcode Reader	Portable Media Player
Trackball	Graphic Tablet
Magnetic Ink Card Reader (MICR)	Optical Character Reader (OCR)
Optical Character Reader (OCR)	Digitizer

An input device is a computer device that allows computer users to enter data into a system and send instructions to the system to execute tasks accordingly. These devices are mainly hardware like keyboards, mice, joysticks, etc. It is the first or primary step in the processing of computer data that is done at the Central Processing Unit (CPU). The delivered signals are received by the CPU which processes them. Input Devices of Computer can be classified further as per the modality like visual or audio, discrete or continuous, and direct or indirect. Some of the main input devices are explained below with their descriptions.

Keyboard
The keyboard is the most fundamental input device of the computer. It is commonly used to insert data on the computer by using keys mounted on a keyboard. It is connected to the computer system through wifi or a USB cable. There are different varieties of keys for different purposes like numerals, letters, special

characters, and functions. It is the main input device to command the computer system.



Mouse

A Mouse is a hand-supported device that enables computer users to move the cursor point on the computer screen. A mouse consists of two buttons namely left and right on its top portion and one trackball at its bottom. It works on a flat surface to select and move the mouse around. The mouse as an input device was invented by Douglas C. Engelbart in 1963.



Joy Stick

Joystick comprises a stick connected to the base at an angle so that it can be easily moved and controlled. It is mainly applied in controlling the movement of characters in video games. It is also employed in the cockpit of an airplane, wheelchairs, cranes, and trucks to regulate them properly. Its function is also to move the cursor on the screen but it is not like a mouse.

Light Pen

A light pen is a pointing input device that is in a pen-like structure. It allows computer users to select options on the screen and even draw on the screen. It

is light-sensitive equipment as photocells are inbuilt in this device that allows the flow of instructions to the CPU. It is generally used with a cathode ray tube (CRT) of the computer.



Microphone

A microphone is an inbuilt voice input device in which different sounds can be collected in their digital form. This input device converts sound instructions into electrical signals. It has to be linked with an amplifier for recording or reproducing the sound.

Scanner

A scanner is an input device that can scan different types of files of different formats like images or text and then convert them into digital signals. It can bring the document files to the computer screen by converting these files into a digital format. It operates like a photocopy machine to make the written information or data on a computer screen from paper.



Barcode Reader

A barcode reader is a type of optical scanner device. It can read and understand the bar code data attached to goods, books, etc. It is like a reading device that throws a ray of light on a bar code and then the aspects and details of that particular item are displayed on the screen.

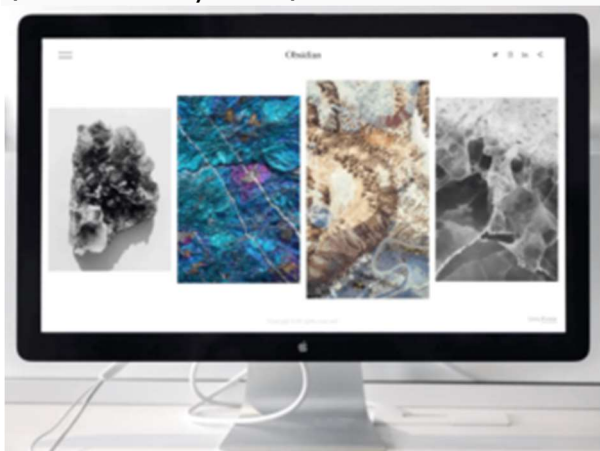
Output Devices of Computer with Examples

An output device is a computer device that displays the result of the inserted input data after getting processed from the CPU. CPU of a computer converts received information from machine language to a human-friendly language and then sends these signals to output devices to execute the tasks as per entered data. These output devices are mainly hardware like printers, monitors, speakers, etc. When the data entered by the input devices is processed by the CPU of the computer then the output devices take care of the result. These results after processing of data may be in image, graphic, textual or audio form. The output devices display the visual contents on the monitor screen while sound contents are delivered to the speaker connected to a computer. Some of the main Output devices of computer are explained below with their descriptions.

Examples of Output Devices of Computer	
Monitor	Speaker
Printer	Projector
Plotter	Braille Reader
Television	Global Positioning System
Headphones	Video Card

Monitor

The monitor is the main output device that displays all the data related to icons, text, images, etc. on its screen. When we enter the command to the computer to execute an action, then the outcome of that action is displayed on the screen of the monitor. Different types of monitors have been developed over time like CRT (Cathode Ray Tube) monitors, Flat-Panel display monitors, etc.



Printer

A Printer is an output device that produces a copy of the pictorial or textual files generally over a page. Its primary function is to print the information on paper. For example, an writer types a complete book on his computer system. He has to take a printout of it in the form of paper so that the book can be reviewed and later published. There are different types of printers in the market for different purposes like impact printers, character printers, line printers, laser printers, etc.



Speakers

A Speaker is an output device that converts electrical instructions into a sound signal. It helps you to listen to sound signals as an outcome of what you enter data into a computer. It is a hardware device that may or may not be attached to the computer system. Now, speakers are becoming wireless devices and can be linked with systems with the help of BlueTooth or else.



Projector

A Projector is the output and optical device that presents visual contents like moving or stationary images on the screen. These projecting devices are generally applied in auditoriums and cinema theatres for screening videos and lighting effects. Once a projector is linked to a computer system then the content displayed on the projection screen will be the same as displayed on the Monitor screen. The difference here is that projector displays contents on the bigger screens.

Headphones

The headphones operate on the same principles as a speaker operates. The only difference between headphones and speaker is the frequency of sound. With the

help of speakers, the released sound can cover a larger area while with the help of headphones, the released sound can cover only a smaller area to make it audible only to a person who is wearing these headphones. These are also called earphones or headsets.



Difference Between Input and Output Devices of Computer

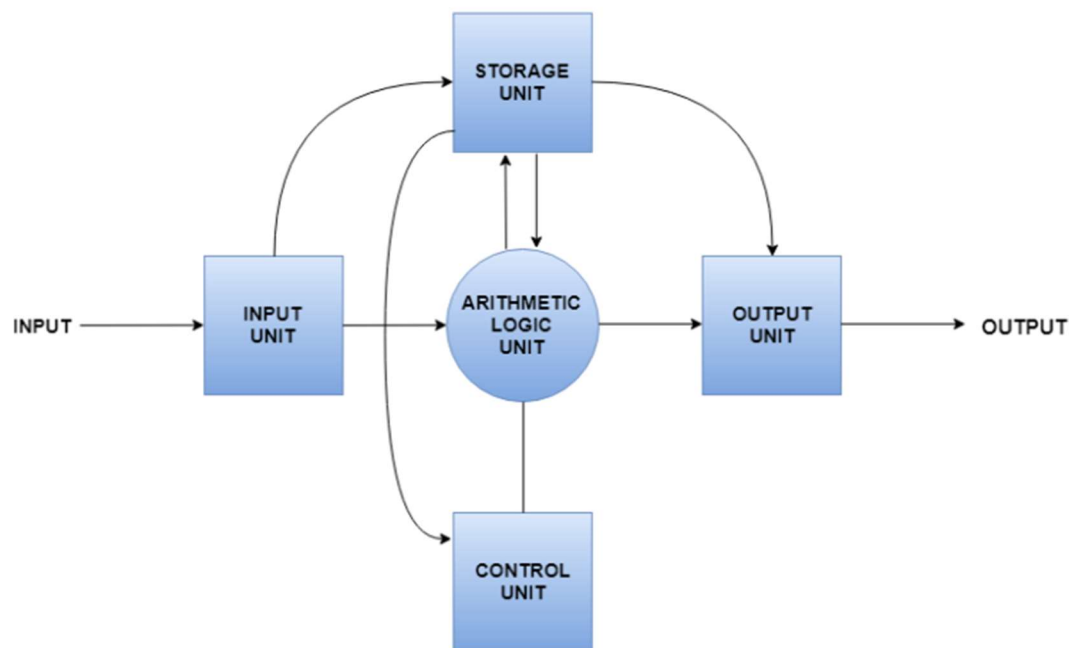
The computer operates smoothly with the help of both input and output devices. CPU plays a vital role in making a bridge between the input and output devices. Here the complete comparison of input and output devices of computers is explained below.

Difference Between Input and Output Devices of Computer	
Input Devices	Output Devices
Accept computer user's data	Reflect computer user's data
The computer users command them	The computer processors command them
Conversion of Human-friendly signals into a machine-friendly language	Conversion of machine-friendly language to user-friendly language
Send signals to the CPU for execution	Send processed signals to the computer user
Assist computer system in receiving data	Assist computer system in displaying data
It is complex in design comparatively	It is lesser complex in design
Example: Keyboard, Scanner, Mouse, etc.	Example: Monitor, Printers, Speaker, etc.,

Computer system Architecture

A computer system is basically a machine that simplifies complicated tasks. It should maximize performance and reduce costs as well as power consumption. The different components in the Computer System Architecture are Input Unit, Output Unit, Storage Unit, Arithmetic Logic Unit, Control Unit etc.

A diagram that shows the flow of data between these units is as follows –



The input data travels from input unit to ALU. Similarly, the computed data travels from ALU to output unit. The data constantly moves from storage unit to ALU and back again. This is because stored data is computed on before being stored again. The control unit controls all the other units as well as their data.

Details about all the computer units are –

- **Input Unit**

The input unit provides data to the computer system from the outside. So, basically it links the external environment with the computer. It takes data from the input devices, converts it into machine language and then loads it into the computer system. Keyboard, mouse etc. are the most commonly used input devices.

- **Output Unit**

The output unit provides the results of computer process to the users i.e it links the computer with the external environment. Most of the

output data is the form of audio or video. The different output devices are monitors, printers, speakers, headphones etc.

- **Storage Unit**

Storage unit contains many computer components that are used to store data. It is traditionally divided into primary storage and secondary storage. Primary storage is also known as the main memory and is the memory directly accessible by the CPU. Secondary or external storage is not directly accessible by the CPU. The data from secondary storage needs to be brought into the primary storage before the CPU can use it. Secondary storage contains a large amount of data permanently.

- **Arithmetic Logic Unit**

All the calculations related to the computer system are performed by the arithmetic logic unit. It can perform operations like addition, subtraction, multiplication, division etc. The control unit transfers data from storage unit to arithmetic logic unit when calculations need to be performed. The arithmetic logic unit and the control unit together form the central processing unit.

- **Control Unit**

This unit controls all the other units of the computer system and so is known as its central nervous system. It transfers data throughout the computer as required including from storage unit to central processing unit and vice versa. The control unit also dictates how the memory, input output devices, arithmetic logic unit etc. should behave.

Computer architecture can be defined as a set of rules and methods that describe the functionality, management and implementation of computers. To be precise, it is nothing but rules by which a system performs and operates.

Sub-divisions

Computer Architecture can be divided into mainly three categories, which are as follows –

- **Instruction set Architecture or ISA** – Whenever an instruction is given to processor, its role is to read and act accordingly. It allocates memory to instructions and also acts upon memory address mode (Direct Addressing mode or Indirect Addressing mode).
- **Micro Architecture** – It describes how a particular processor will handle and implement instructions from ISA.
- **System design** – It includes the other entire hardware component within the system such as virtualization, multiprocessing.

Role of computer Architecture

The main role of Computer Architecture is to balance the performance, efficiency, cost and reliability of a computer system.

For Example - Instruction set architecture (ISA) acts as a bridge between computer's software and hardware. It works as a programmer's view of a machine. Computers can only understand binary language (i.e., 0, 1) and users understand high level language (i.e., if else, while, conditions, etc). So to communicate between user and computer, Instruction set Architecture plays a major role here, translating high level language to binary language.

Computer Network Architecture

A network model represents the organization of multiple computers in a network. It describes how individual computers are interconnected in a network.

The computer network architecture is as follows –

Centralized Computing Architecture

In centralized computing architecture, one powerful computer is used to serve one or more low powered computers. In the centralized model, the nodes are not connected; they are only connected to the server.

The centralized computing architecture contains the following –

- All processing holds a function in the central, mainframe computer.
- Terminals are linked to the central computer and operation just as input/output devices.
- Networks can be employed to interconnect at least two mainframe computers. Terminals relate only to the mainframe, never to one another.

Distributed Computing Architecture

Distributed architecture interconnects one or more personal computers known as nodes. It allows various features such as file sharing, hardware sharing or network sharing. In the distributed model, the nodes can handle their data and depend on the network for administration other than data processing.

For example, a network of Window 2000 can have a computer that is regarded as a Print server. The print server would handle all the printing jobs of the network.

Distributed computing architecture contains the following –

- Various computers are efficient in performing independently
- Tasks are finished locally on multiple computers.
- Networks allow the computers to transfer data and service but do not support processing assistance.

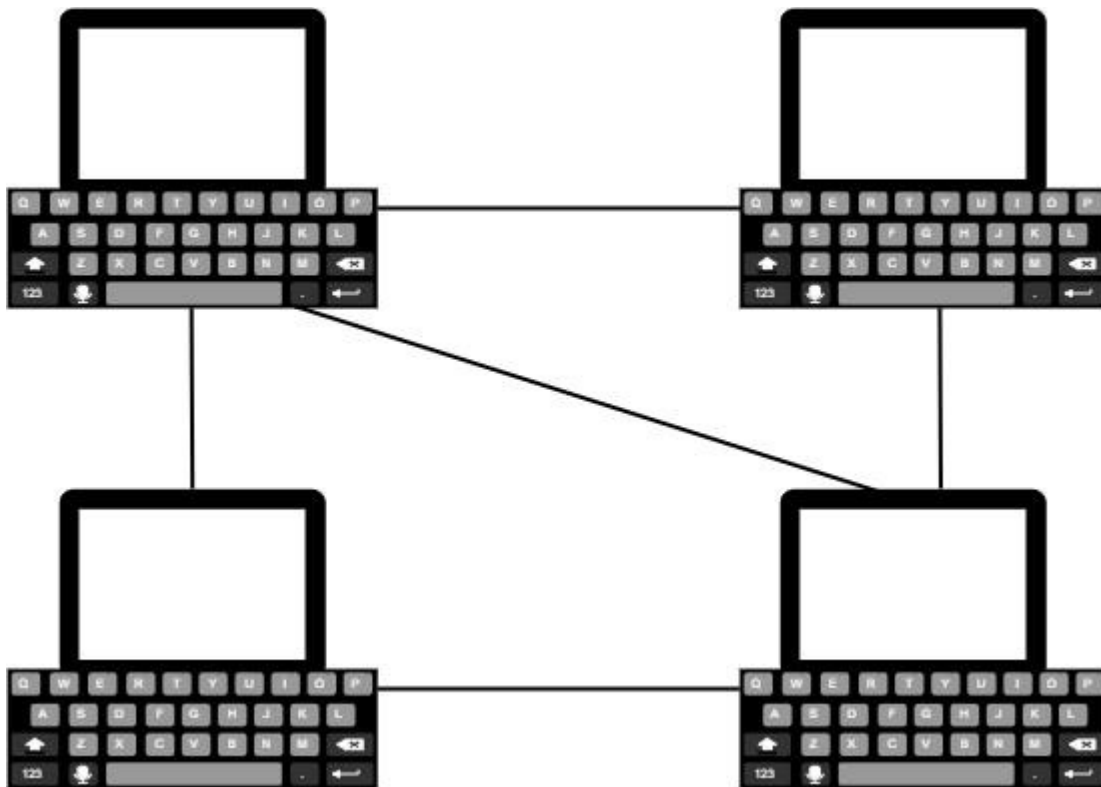
Collaborative Computing Architecture

The collaborative computing architecture is a mixture of centralized and distributed computing. In the collaborative model, the individual members of a network can process their user's basic need.

For example, a database server such as MSSQL server or ORACLE server sees or handles all the database related processing of all the nodes of networks. But requests other than the database will be processed by the model.

Peer-to-Peer Architecture

In this architecture, there are no dedicated servers. All computers are similar and, therefore, are termed as a peer. Each of these machines functions both as a user and a server. This arrangement is applicable for environments with a finite number of clients (usually ten or less).



Advantages

- It is accessible to design and perpetuation.
- The network is not dependent on a precise computer.
- Linking new computers in a peer to peer network is significantly more straightforward.

What Do Computer Networks Do?

Computer Networks are one of the important aspects of Computer Science. In the early days, it is used for data transmission on telephone lines and had a very limited use, but nowadays, it is used in a variety of places.

Computer Networks help in providing better connectivity that helps nowadays. Modern computer networks have the following functionality like

- Computer Networks help in operating virtually.
- Computer Networks integrate on a large scale.
- Computer Networks respond very quickly in case of conditions change.
- Computer Networks help in providing data security.

Criteria of a Good Network

1. **Performance:** It can be measured in many ways, including transmit time and response time. Transit time is the amount of time required for a message to travel from one device to another. Response time is the elapsed time between an inquiry and a response. The performance of the network depends on a number of factors, including the number of users, the type of medium & Hardware

2. **Reliability:** In addition to accuracy is measured by frequency of failure, the time it takes a link to recover from failure, and the network's robustness in catastrophe.
3. **Security:** Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data loss.

Goals of Computer Networking

- Programs do not have to execute on a single system because of resource and load sharing.
- Reduced costs – Multiple machines can share printers, tape drives, and other peripherals.
- Reliability – If one machine fails, another can take its place.
- Scalability (it's simple to add more processors or computers)
- Communication and mail (people living apart can work together)
- Information Access (remote information access, access to the internet, e-mail, video conferencing, and online shopping)
- Entertainment that is interactive (online games, videos, etc.)
- Social Networking

Types of Computer Networks

Division Based on the Communication Medium

- **Wired Network:** As we all know, "wired" refers to any physical medium made up of cables. Copper wire, twisted pair, or fiber optic cables are all options. A wired network employs wires to link devices to the Internet or another network, such as laptops or desktop PCs.
- **Wireless Network:** "Wireless" means without wire, media that is made up of electromagnetic waves (EM Waves) or infrared waves. Antennas or sensors will be present on all wireless devices. Cellular phones, wireless sensors, TV remotes, satellite dish receivers, and laptops with WLAN cards are all examples of wireless devices. For data or voice communication, a wireless network uses radio frequency waves rather than wires.

Division Based on Area Covered

- **Local Area Network (LAN):** A LAN is a network that covers an area of around 10 kilometers. For example, a college network or an office network. Depending upon the needs of the organization, a LAN can be a single office, building, or Campus. We can have two PCs and one printer in-home office or it can extend throughout the company and include audio and video devices. Each host in LAN has an identifier, an address that defines hosts in LAN. A packet sent by the host to another host carries both the source host's and the destination host's address.
- **Metropolitan Area Network (MAN):** MAN refers to a network that covers an entire city. For example: consider the cable television network.

- **Wide Area Network (WAN):** WAN refers to a network that connects countries or continents. For example, the Internet allows users to access a distributed system called www from anywhere around the globe. WAN interconnects connecting devices such as switches, routers, or modems. A LAN is normally privately owned by an organization that uses it. We see two distinct examples of WANs today: point-to-point WANs and Switched WANs
 - **Point To Point:** Connects two connecting devices through transmission media.
 - **Switched:** A switched WAN is a network with more than two ends.

Based on Types of Communication

- **Point To Point networks:** Point-to-Point networking is a type of data networking that establishes a direct link between two networking nodes. A direct link between two devices, such as a computer and a printer, is known as a point-to-point connection.
- **Multipoint:** is the one in which more than two specific devices share links. In the multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a spatially shared connection.
- **Broadcast networks:** In broadcast networks, a signal method in which numerous parties can hear a single sender. Radio stations are an excellent illustration of the "Broadcast Network" in everyday life. The radio station is a sender of data/signal in this scenario, and data is only intended to travel in one direction. Away from the radio transmission tower, to be precise.

Based on the Type of Architecture

- **P2P Networks:** Computers with similar capabilities and configurations are referred to as peers. "Peer to Peer" is the abbreviation for "peer to peer." The "peers" in a peer-to-peer network are computer systems that are connected to each other over the Internet. Without the use of a central server, files can be shared directly between systems on the network.
- **Client-Server Networks:** Each computer or process on the network is either a client or a server in a client-server architecture (client/server). The client asks for services from the server, which the server provides. Servers are high-performance computers or processes that manage disc drives (file servers), printers (print servers), or network traffic (network servers)
- **Hybrid Networks:** The hybrid model refers to a network that uses a combination of client-server and peer-to-peer architecture. Eg: Torrent.

Types of Computer Network Architecture

Computer Network Architecture is of two types. These types are mentioned below.

1. Client-Server Architecture: It is basically the architecture where the clients and the server are connected as two clients can communicate with each other and the devices present work as servers in the network.

2. Peer-to-Peer Architecture: computers are connected to each other and each computer is equally capable of working as there is no central server here. Each device present here can be used as a client or server.

Types of Enterprise Computer Networks

There are three main types of Enterprise Computer Networks which are mentioned below.

1. Local Area Network (LAN): These are small-scale networks used in small companies or as test networks. It has a limited size.

2. Wide Area Networks (WAN): These are networks that are used for a larger area than local area networks and are used for long-distance communication.

3. Service Provider Networks: Service Provider Networks are the networks that help in wireless communication, high-speed internet access, etc.

Network Topology

Bus Topology

Every computer and network device is connected to a single cable in a bus topology network. Linear Bus topology is defined as having exactly two terminals.

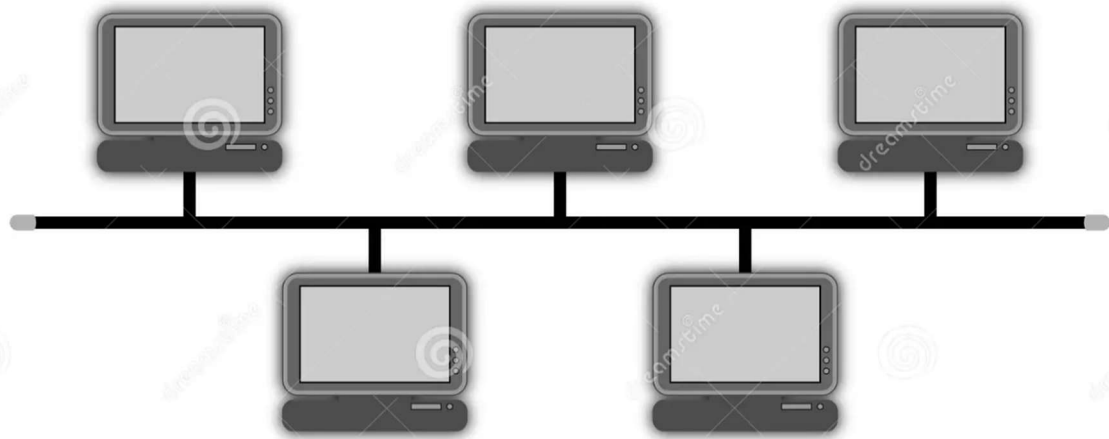
Advantages

- Installation is simple.
- Compared to mesh, star, and tree topologies, the bus utilizes less cabling.

Disadvantages

- Difficulty in reconfiguring and isolating faults.
- A bus cable malfunction or break interrupts all communication.

Bus Topology



Bus Topology

Ring Topology

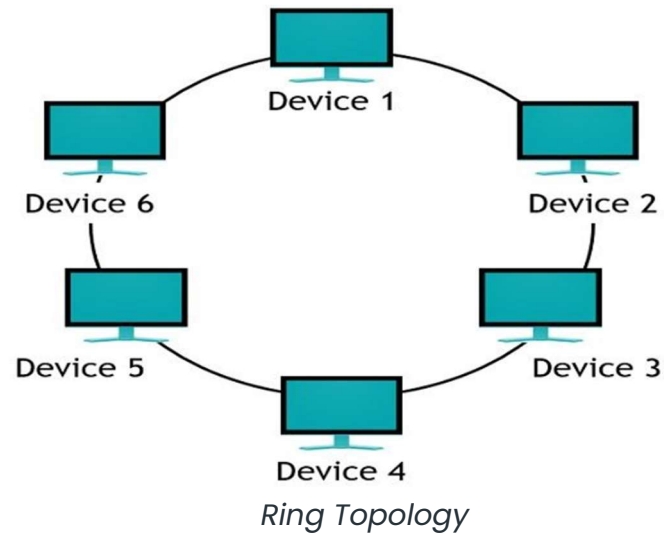
The topology is named ring topology because one computer is connected to another, with the final one being connected to the first. Exactly two neighbors for each device. A signal is passed along the ring in one direction. Each ring incorporates a repeater.

Advantages

- Data transmission is relatively straightforward because packets only move in one direction.
- There is no requirement for a central controller to manage communication between nodes.
- Easy installation & Reconfiguration
- Simplified Faulty connections

Disadvantages

- In a Unidirectional Ring, a data packet must traverse through all nodes.
- All computers must be turned on in order for them to connect with one another.



Star Topology

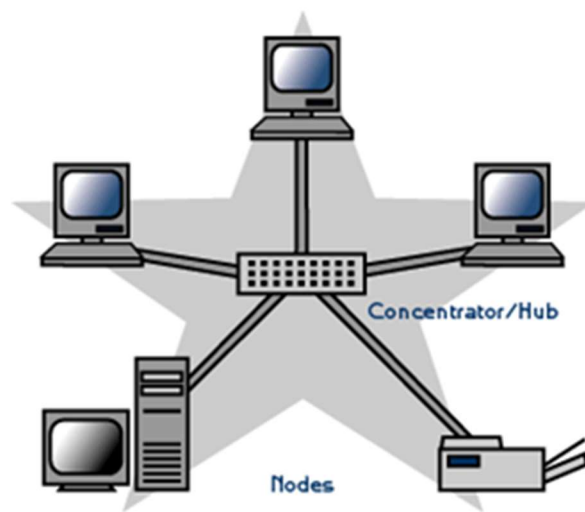
Each device in a star topology has a dedicated point-to-point link to a central controller, which is commonly referred to as the HUB. There is no direct connection between the devices. Traffic between the devices is not allowed in this topology. As an exchange, the controller is used.

Advantages

- When attaching or disconnecting devices, there are no network interruptions.
- It's simple to set up and configure.
- Identifying and isolating faults is simple.
- Less Expensive than mesh
- Easy to install & configure

Disadvantages

- Nodes attached to the hub, switch, or concentrator is failed if they fail.
- Because of the expense of the hubs, it is more expensive than linear bus topologies.
- More cable is required compared to a bus or ring
- Too much dependency on Hub



Star Topology

Mesh Topology

Every device in a mesh topology has dedicated point-to-point connectivity to every other device. The term “dedicated” refers to the fact that the link exclusively transports data between the two devices it links. To connect n devices, a fully connected mesh network contains $n * (n-1) / 2$ physical channels.

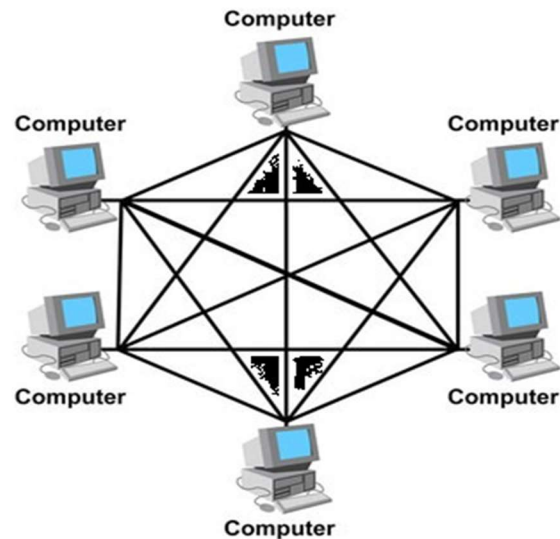
Advantages

- Data can be sent from multiple devices at the same time. This topology can handle a lot of traffic.
- Even if one of the connections fails, a backup is always available. As a result, data transit is unaffected.
- Physical boundaries prevent other users from gaining access to messages
- Point to Point links make fault transmission & fault isolation easy

Disadvantages

- The amount of cabling and the number of I/O ports that are necessary.
- The sheer bulk of wiring can be greater than the available space can accommodate.
- It is difficult to install and reconfigure.

Example: connection of telephone regional office in which each regional office needs to be connected to every other regional office.



Mesh Topology

Tree Topology

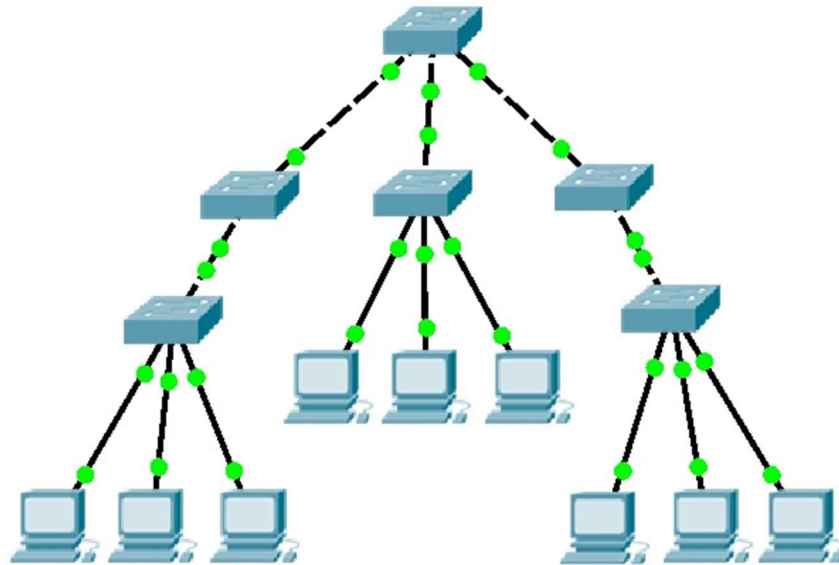
The topology of a tree is similar to that of a star. Nodes in a tree, like those in a star, are connected to a central hub that manages network traffic. It has a root node, which is connected to all other nodes, producing a hierarchy. Hierarchical topology is another name for it. The number of Star networks is connected via Bus in Tree Topology.

Advantages

- Network expansion is both possible and simple.
- We partition the entire network into pieces (star networks) that are easier to manage and maintain.
- Other segments are unaffected if one segment is damaged.

Disadvantages

- Tree topology relies largely on the main bus cable because of its basic structure, and if it fails, the entire network is handicapped.
- Maintenance becomes more challenging when more nodes and segments are added.



Tree Topology

Networking Devices

Basic hardware interconnecting network nodes, such as Network Interface Cards (NICs), Bridges, Hubs, Switches, and Routers, are used in all networks. In addition, a mechanism for connecting these building parts is necessary, which is usually galvanic cable and optical cable are less popular ("optical fiber") The following are the network devices :

- **NIC (Network Interface Card):** A network card, often known as a network adapter or NIC (network interface card), is computer hardware that enables computers to communicate via a network. It offers physical access to networking media and, in many cases, MAC addresses serve as a low-level addressing scheme. Each network interface card has a distinct identifier. This is stored on a chip that is attached to the card.
- **Repeater:** A repeater is an electrical device that receives a signal, cleans it of unwanted noise, regenerates it, and retransmits it at a higher power level or to the opposite side of an obstruction, allowing the signal to travel greater distances without degradation. In the majority of twisted pair Ethernet networks, Repeaters are necessary for cable lengths longer than 100 meters in some systems. Repeaters are based on physics.
- **Hub:** A hub is a device that joins together many twisted pairs or fiber optic Ethernet devices to give the illusion of a formation of a single network segment. The device can be visualized as a multiport repeater. A network hub is a relatively simple broadcast device. Any packet entering any port is regenerated and broadcast out on all other ports, and hubs do not control any of the traffic that passes through them. Packet collisions occur as a result of every packet being sent out through

all other ports, substantially impeding the smooth flow of communication.

- **Bridges:** Bridges broadcast data to all the ports but not to the one that received the transmission. Bridges, on the other hand, learn which MAC addresses are reachable through specific ports rather than copying messages to all ports as hubs do. Once a port and an address are associated, the bridge will only transport traffic from that address to that port.
- **Switches:** A switch differs from a hub in that it only forwards frames to the ports that are participating in the communication, rather than all of the ports that are connected. The collision domain is broken by a switch, yet the switch depicts itself as a broadcast domain. Frame-forwarding decisions are made by switches based on MAC addresses.
- **Routers:** are networking devices that use headers and forwarding tables to find the optimal way to forward data packets between networks. A router is a computer networking device that links two or more computer networks and selectively exchanges data packets between them. A router can use address information in each data packet to determine if the source and destination are on the same network or if the data packet has to be transported between networks. When numerous routers are deployed in a wide collection of interconnected networks, the routers share target system addresses so that each router can develop a table displaying the preferred pathways between any two systems on the associated networks.
- **Gateways:** To provide system compatibility, a gateway may contain devices such as protocol translators, impedance-matching devices, rate converters, fault isolators, or signal translators. It also necessitates the development of administrative procedures that are acceptable to both networks. By completing the necessary protocol conversions, a protocol translation/mapping gateway joins networks that use distinct network protocol technologies.

Internet

This is a larger network that allows computer networks controlled by enterprises, governments, colleges, and other organizations all over the world to communicate with one another. As a result, there is a tangle of cables, computers, data centers, routers, servers, repeaters, satellites, and Wi-Fi towers that allow digital data to go around the world.

The Internet is a vast network of networks that functions as a networking infrastructure. It links millions of computers throughout the world, creating a network in which any computer can talk with any other computer as long as they are both linked to the Internet.

The Internet is a global network of interconnected computers that communicate and share information using a standardized Internet Protocol Suite.

Software and its Types

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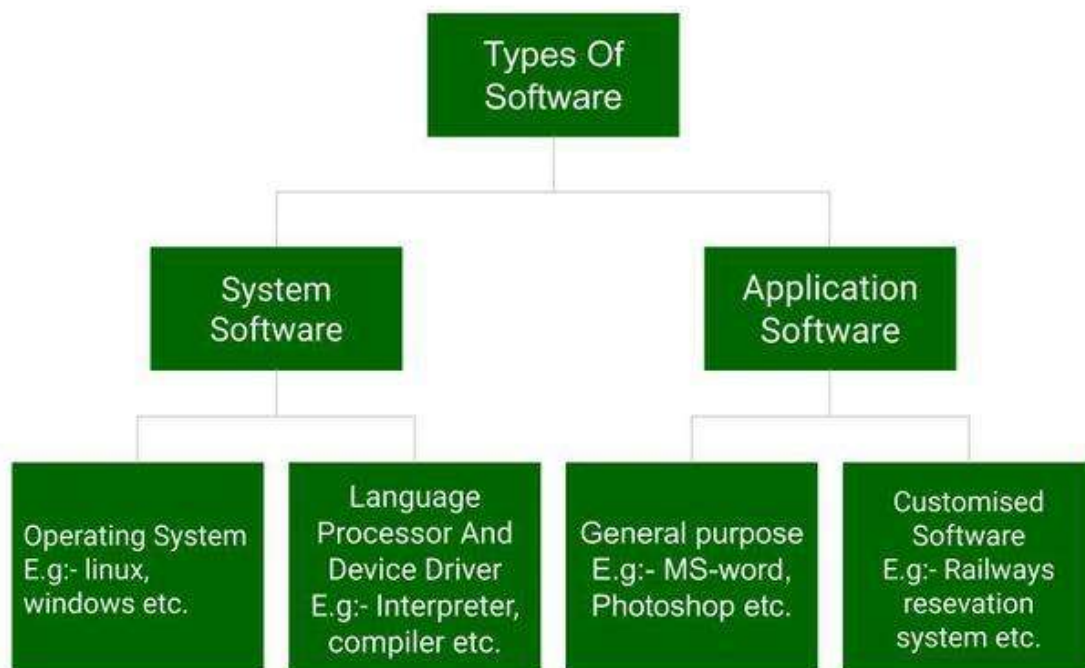
Software is a collection of instructions, data, or computer programs that are used to run machines and carry out particular activities. It is the antithesis of hardware, which refers to a computer's external components. A device's running programs, scripts, and applications are collectively referred to as "software" in this context.

What is Software?

In a Computer system, the software is basically a set of instructions or commands that tell a computer what to do. In other words, the software is a computer program that provides a set of instructions to execute a user's commands and tell the computer what to do. For example like MS-Word, MS-Excel, MS-Powerpoint, etc.

Types of Software

It is a collection of data that is given to the computer to complete a particular task. The chart below describes the types of software:



Above is the diagram of types of software. Now we will briefly describe each type and its subtypes:

1. **System Software**

- Operating System

- Language Processor
 - Device Driver
2. **Application Software**
- General Purpose Software
 - Customize Software
 - Utility Software

System Software

System Software is software that directly operates the computer hardware and provides the basic functionality to the users as well as to the other software to operate smoothly. Or in other words, system software basically controls a computer's internal functioning and also controls hardware devices such as monitors, printers, and storage devices, etc. It is like an interface between hardware and user applications, it helps them to communicate with each other because hardware understands machine language (i.e. 1 or 0) whereas user applications work in human-readable languages like English, Hindi, German, etc. so system software converts the human-readable language into machine language and vice versa.

Types of System Software

It has two subtypes which are:

1. **Operating System:** It is the main program of a computer system. When the computer system is ON it is the first software that loads into the computer's memory. Basically, it manages all the resources such as computer memory, CPU, Printer, hard disk, etc., and provides an interface to the user, which helps the user to interact with the computer system. It also provides various services to other computer software. Examples of operating systems are Linux, Apple macOS, Microsoft Windows, etc.
2. **Language Processor:** As we know that system software converts the human-readable language into a machine language and vice versa. So, the conversion is done by the language processor. It converts programs written in high-level programming languages like Java, C, C++, Python etc (known as source code), into sets of instructions that are easily readable by machines (known as object code or machine code).
3. **Device Driver:** A device driver is a program or software that controls a device and helps that device to perform its functions. Every device like a printer, mouse, modem, etc. needs a driver to connect with the computer system eternally. So, when you connect a new device with your computer system, first you need to install the driver of that device so that your operating system knows how to control or manage that device.

Features of System Software

Let us discuss some of the features of System Software:

- System Software is closer to the computer system.
- System Software is written in a low-level language in general.
- System software is difficult to design and understand.
- System software is fast in speed(working speed).
- System software is less interactive for the users in comparison to application software.

Application Software

Software that performs special functions or provides functions that are much more than the basic operation of the computer is known as application software. Or in other words, application software is designed to perform a specific task for end-users. It is a product or a program that is designed only to fulfill end-users' requirements. It includes word processors, spreadsheets, database management, inventory, payroll programs, etc.

Types of Application Software

There are different types of application software and those are:

1. **General Purpose Software:** This type of application software is used for a variety of tasks and it is not limited to performing a specific task only. For example, MS-Word, MS-Excel, PowerPoint, etc.
2. **Customized Software:** This type of application software is used or designed to perform specific tasks or functions or designed for specific organizations. For example, railway reservation system , airline reservation system, invoice management system, etc.
3. **Utility Software:** This type of application software is used to support the computer infrastructure. It is designed to analyze, configure, optimize and maintains the system, and take care of its requirements as well. For example, antivirus , disk fragmenter, memory tester, disk repair, disk cleaners, registry cleaners, disk space analyzer, etc.

Features of Application Software

Let us discuss some of the features of Application Software:

- An important feature of application software is it performs more specialized tasks like word processing, spreadsheets, email, etc.
- Mostly, the size of the software is big, so it requires more storage space.
- Application software is more interactive for the users, so it is easy to use and design.
- The application software is easy to design and understand.
- Application software is written in a high-level language in general.

Difference Between System Software and Application Software

Now, let us discuss some difference between system software and application software:

System Software	Application Software
It is designed to manage the resources of the computer system, like memory and process management, etc.	It is designed to fulfil the requirements of the user for performing specific tasks.
Written in a low-level language.	Written in a high-level language.
Less interactive for the users.	More interactive for the users.
System software plays vital role for the effective functioning of a system.	Application software is not so important for the functioning of the system, as it is task specific.
It is independent of the application software to run.	It needs system software to run.

Software and Its Type – FAQs

1. What is difference between a software and a program?

Both are for enabling the computer to perform specific task. The software is the collection of programs. we have create a program to perform the task and they compiled if there is no error in program then program sent to create a software. Program is the set of instructions that are written by programmer in a language while the software is a collection of program that will enable the system to perform the specific task.

2. What is an example of Software?

You need software in order for the computer to operate effectively. It is a collection of data that is given to the computer to complete a particular task. three main categories of software are application software, programming software, and system software. All these three are important for the particular task to be performed.

Below are some software examples.

- *Management Tools*
- *Compression Tools*

- Internet Explorer
- ERP(Enterprise Resource Planning)
- BI (Business Intelligence)
- Customer Support Systems

3. Which software is used to control the operations of a computer?

There are two types of software system software and application software. System Software is used to control the operations and also controls a computer's internal functioning and hardware devices.

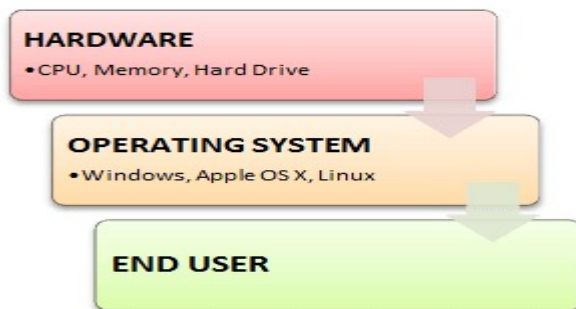
4. Which software is designed to solve a specific problem or to do a specific task

Application Software is designed to solve a specific problem or to do a specific task Because a software that performs special functions or provides function which are much more than basic operation of the computer are application software.

Operating System

An **Operating System (OS)** is a software that acts as an interface between computer hardware components and the user. Every computer system must have at least one operating system to run other programs. Applications like Browsers, MS Office, Notepad Games, etc., need some environment to run and perform its tasks.

The OS helps you to communicate with the computer without knowing how to speak the computer's language. It is not possible for the user to use any computer or mobile device without having an operating system.



History Of OS

- Operating systems were first developed in the late 1950s to manage tape storage
- The General Motors Research Lab implemented the first OS in the early 1950s for their IBM 701
- In the mid-1960s, operating systems started to use disks
- In the late 1960s, the first version of the Unix OS was developed
- The first OS built by Microsoft was DOS. It was built in 1981 by purchasing the 86-DOS software from a Seattle company
- The present-day popular OS Windows first came to existence in 1985 when a GUI was created and paired with MS-DOS.

Types of Operating System (OS)

Following are the popular types of OS (Operating System):

- Batch Operating System
- Multitasking/Time Sharing OS
- Multiprocessing OS
- Real Time OS
- Distributed OS
- Network OS
- Mobile OS

Batch Operating System

Some computer processes are very lengthy and time-consuming. To speed the same process, a job with a similar type of needs are batched together and run as a group.

The user of a batch operating system never directly interacts with the computer. In this type of OS, every user prepares his or her job on an offline device like a punch card and submit it to the computer operator.

Multi-Tasking/Time-sharing Operating systems

Time-sharing operating system enables people located at a different terminal(shell) to use a single computer system at the same time. The processor time (CPU) which is shared among multiple users is termed as time sharing.

Real time OS

A real time operating system allots time interval to process and respond to inputs which is very small. Examples: Military Software Systems, Space Software Systems are the Real time OS example.

Distributed Operating System

Distributed systems use many processors located in different machines to provide very fast computation to its users.

Network Operating System

Network Operating System runs on a server. It provides the capability to serve to manage data, user, groups, security, application, and other networking functions.

Mobile OS

Mobile operating systems are those OS which is especially that are designed to power smartphones, tablets, and wearables devices.

Some most famous mobile operating systems are Android and iOS, but others include BlackBerry, Web, and watchOS.

Functions of Operating System

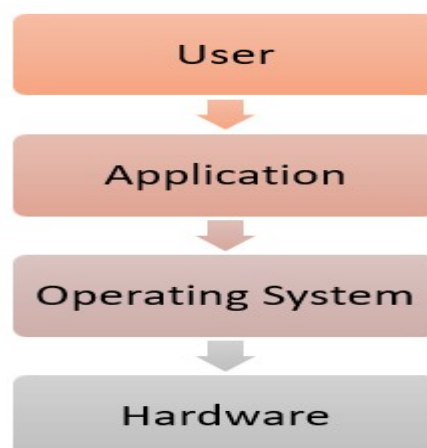
Some typical operating system functions may include managing memory, files, processes, I/O system & devices, security, etc.

Below are the main functions of Operating System:



Functions of Operating System

1. **Process management:** Process management helps OS to create and delete processes. It also provides mechanisms for synchronization and communication among processes.
2. **Memory management:** Memory management module performs the task of allocation and deallocation of memory space to programs in need of this resources.
3. **File management:** It manages all the file-related activities such as organization storage, retrieval, naming, sharing, and protection of files.
4. **Device Management:** Device management keeps tracks of all devices. This module also responsible for this task is known as the I/O controller. It also performs the task of allocation and deallocation of the devices.
5. **I/O System Management:** One of the main objects of any OS is to hide the peculiarities of that hardware devices from the user.
6. **Secondary-Storage Management:** Systems have several levels of storage which includes primary storage, secondary storage, and cache storage. Instructions and data must be stored in primary storage or cache so that a running program can reference it.
7. **Security:** Security module protects the data and information of a computer system against malware threat and authorized access.
8. **Command interpretation:** This module is interpreting commands given by the and acting system resources to process that commands.
9. **Networking:** A distributed system is a group of processors which do not share memory, hardware devices, or a clock. The processors communicate with one another through the network.
10. **Job accounting:** Keeping track of time & resource used by various job and users.
11. **Communication management:** Coordination and assignment of compilers, interpreters, and another software resource of the various users of the computer systems.

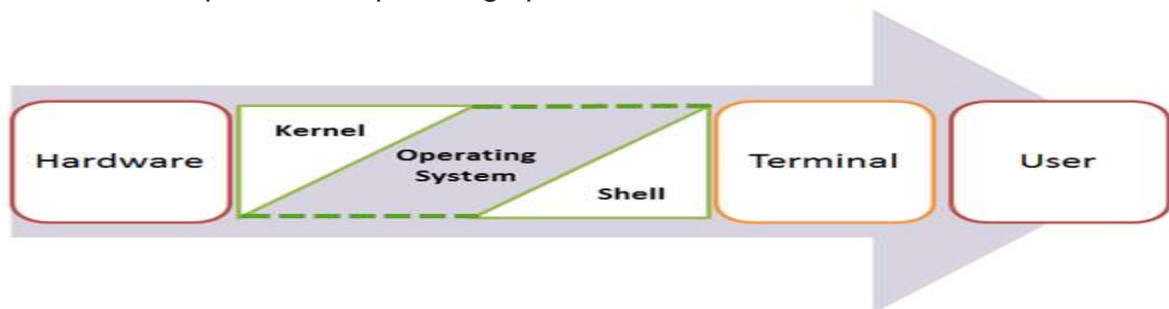


Advantage of Operating System

- Allows you to hide details of hardware by creating an abstraction
- Easy to use with a GUI
- Offers an environment in which a user may execute programs/applications
- The operating system must make sure that the computer system convenient to use
- Operating System acts as an intermediary among applications and the hardware components
- It provides the computer system resources with easy to use format
- Acts as an intermediary between all hardware and software of the system

What is Kernel in Operating Systems?

The kernel is the central component of a computer operating system. The only job performed by the kernel is to manage the communication between the software and the hardware. A Kernel is at the nucleus of a computer. It makes the communication between the hardware and software possible. While the Kernel is the innermost part of an operating system, a shell is the outermost one.



Major Operating systems :

Name	Features
MS-DOS	stands for Microsoft Disk Operating system. does not take much space for installation (about 8MB). gives more control to the processes as it has a command-line user interface. single-user, single-tasking operating system. has limited graphics features. not compatible with current browsers and the Internet.
Unix	was developed in 1969 by Ken Thompson, Dennis Ritchie and others at AT & T Bell Laboratories. was earlier known as UNICS [Uniplexed Information and Computing Service]. Was rewritten in C language. This made the OS portable.

	one may use either the CUI that gives more control or flexibility or GUI which makes it simple. very flexible and can be installed on micro , mini , mainframe and supercomputers. very reliable, secured and robust.
Linux	developed by Linus Torvalds in 1992. Unix-like OS and is available free. like DOS, Linux is a command-line user interface. It also gives GUI interfaces called desktop environments like GNOME and K Desktop Environment (KDE). multi-user, multi-tasking OS. reliable and secured OS.
Windows	first version of Windows was developed by Microsoft Corporation in 1985. has a graphical user interface and is thus user-friendly. Multi-tasking OS. supports digital audio and video , CD and DVD drives . contains built-in networking, which allows users to share files and applications with each other, if the computers are connected in a network. can support a variety of application software.
MAC	It is a series of graphical user interface-based OS developed by Apple Inc. for their Macintosh line of computer systems. extremely visually appealing. multi-user, multi-tasking OS. requires a low level of maintenance with fewer occurrence of worms, viruses and spyware.

Mobile Operating Systems :

Name	Features
Android	OS for mobile devices such as smart phones and tablet PCs. developed by the Open Handset Alliance led by Google. based on the Linux kernel and most applications are written in Java. has an integrated browser based on the open source Webkit engine. gives media support for common audio , video and image formats. Was purchased by google in 2005.
Symbian	OS for mobile devices such as smartphones.

	originally developed by Symbian Ltd but is currently maintained by Nokia. has a graphics toolkit known as AVKON. was the first mobile platform to make use of WebKit browser. most applications are written in C++, though Symbian devices can be programmed using Python , Java ME , Flash Lite , .NET and so on.
BlackBerry	It is a proprietary mobile OS. has been developed by Research In Motion for the BlackBerry line of smartphones. multitasking mobile OS. supports specialised input devices like trackwheel , trackball , trackpad, and touchscreen. BlackBerry platform is best known for its native support for corporate email.
Windows Mobile	has been developed by Microsoft. Internet Explorer Mobile is the default browser. contains Office Mobile- a suite of mobile versions of Microsoft Office. based on Windows CE 5.2 kernel. Internet Explorer Mobile 11 is the browser now used in Windows Mobile and new browser Microsoft Edge replaced IE in Windows 10 mobile.

Computer Storage

The storage unit is a part of the computer system which is employed to store the information and instructions to be processed. A storage device is an integral part of the computer hardware which stores information/data to process the result of any computational work. Without a storage device, a computer would not be able to run or even boot up. Or in other words, we can say that a storage device is hardware that is used for storing, porting, or extracting data files. It can also store information/data both temporarily and permanently.

Types of Computer Memory

1. Primary Memory
2. Secondary Memory
3. Tertiary Memory

1. Primary Memory: It is also known as internal memory and main memory. This is a section of the CPU that holds program instructions, input data, and intermediate results. It is generally smaller in size. RAM (Random Access Memory) and ROM (Read Only Memory) are examples of primary storage.

2. Secondary Memory: Secondary storage is a memory that is stored external to the computer. It is mainly used for the permanent and long-term storage of programs and data. Hard Disks, CDs, DVDs, Pen/Flash drives, SSD, etc, are examples of secondary storage.

3. Tertiary Memory: Tertiary Memory is a type of Memory that is rarely used in personal computers and due to this, tertiary memory is not considered to be an important one. Tertiary memory works automatically without human intervention.

Types of Computer Storage Devices

Now we will discuss different types of storage devices available in the market. These storage devices have their own specification and use. Some of the commonly used storage devices are:

1. Primary Storage Devices
2. Magnetic Storage Devices
3. Flash memory Devices
4. Optical Storage Devices
5. Cloud and Virtual Storage

1. Primary Storage Devices

- **RAM:** It stands for Random Access Memory. It is used to store information that is used immediately or we can say that it is a temporary memory. Computers bring the software installed on a hard disk to RAM to process it and to be used by the user. Once, the computer is turned off, the data is deleted. With the help of RAM, computers can perform multiple tasks like loading applications, browsing the web, editing a spreadsheet, experiencing the newest game, etc. It allows you to modify quickly among these tasks, remembering where you're in one task once you switch to a

different task. It is also used to load and run applications, like your spreadsheet program, answers commands, like all edits you made within the spreadsheet, or toggle between multiple programs, like once you left the spreadsheet to see the email. Memory is nearly always actively employed by your computer. It ranges from 1GB – 32GB/64GB depending upon the specifications. There are different types of RAM, and although they all serve the same purpose, the most common ones are :

- **SRAM:** It stands for Static Random Access Memory. It consists of circuits that retain stored information as long as the power supply is on. It is also known as volatile memory. It is used to build Cache memory. The access time of SRAM is lower and it is much faster as compared to DRAM but in terms of cost, it is costly as compared to DRAM.
- **DRAM:** It stands for Dynamic Random Access Memory. It is used to store binary bits in the form of electrical charges that are applied to capacitors. The access time of DRAM is slower as compared to SRAM but it is cheaper than SRAM and has a high packaging density.
- **SDRAM:** It stands for Synchronous Random Access Memory. It is faster than DRAM. It is widely used in computers and others. After SDRAM was introduced, the upgraded version of double data rate RAM, i.e., DDR1, DDR2, DDR3, and DDR4 was entered into the market and widely used in home/office desktops and laptops.
- **ROM:** It stands for Read Only Memory. The data written or stored in these devices are non-volatile, i.e, once the data is stored in the memory cannot be modified or deleted. The memory from which will only read but cannot write it. This type of memory is non-volatile. The information is stored permanently during manufacture only once. ROM stores instructions that are used to start a computer. This operation is referred to as bootstrap. It is also used in other electronic items like washers and microwaves. ROM chips can only store a few megabytes (MB) of data, which ranges between 4 and 8 MB per ROM chip. There are two types of ROM:
 - **PROM:** PROM is Programmable Read Only Memory. These are ROMs that can be programmed. A special PROM programmer is employed to enter the program on the PROM. Once the chip has been programmed, information on the PROM can't be altered. PROM is non-volatile, that is data is not lost when power is switched off.
 - **EPROM:** Another sort of memory is the Erasable Programmable Read Only Memory. It is possible to erase the info which has

been previously stored on an EPROM and write new data onto the chip.

- **EEPROM:** EEPROM is Electrically Erasable Read Only Memory. Here, data can be erased without using ultraviolet light, with the use of just applying the electric field.

2. Magnetic Storage Devices

- **Floppy Disk:** Floppy Disk is also known as a floppy diskette. It is generally used on a personal computer to store data externally. A Floppy disk is made up of a plastic cartridge and secured with a protective case. Nowadays floppy disk is replaced by new and effective storage devices like USB, etc.
- **Hard Disk:** Hard Disk is a storage device (HDD) that stores and retrieves data using magnetic storage. It is a non-volatile storage device that can be modified or deleted a number of times without any problem. Most computers and laptops have HDDs as their secondary storage device. It is actually a set of stacked disks, just like phonograph records. In every hard disk, the data is recorded electromagnetically in concentric circles or we can say track present on the hard disk, and with the help of a head just like a phonograph arm (but fixed in a position) to read the information present on the track. The read-write speed of HDDs is not so fast but decent. It ranges from a few GBs to a few and more TB.
- **Magnetic Card:** It is a card in which data is stored by modifying or rearranging the magnetism of tiny iron-based magnetic particles present on the band of the card. It is also known as a swipe card. It is used like a passcode (to enter the house or hotel room), credit card, identity card, etc.
- **Tape Cassette:** It is also known as a music cassette. It is a rectangular flat container in which the data is stored in an analog magnetic tape. It is generally used to store audio recordings.
- **SuperDisk:** It is also called LS-240 and LS-120. It is introduced by Imation Corporation and it is popular with OEM computers. It can store data up to 240 MB.

3. Flash Memory Devices

It is a cheaper and more portable storage device. It is the most commonly used device to store data because it is more reliable and efficient as compared to other storage devices. Some of the commonly used flash memory devices are:

- **Pen Drive:** It is also known as a USB flash drive that includes flash memory with an integrated USB interface. We can directly connect these devices to our computers and laptops and read/write data into them in a much

faster and more efficient way. These devices are very portable. It ranges from 1GB to 256GB generally.

- **SSD:** It stands for Solid State drive, a mass storage device like HDD. It is more durable because it does not contain optical disks inside like hard disks. It needs less power as compared to hard disks, is lightweight, and has 10x faster read and writes speed as compared to hard disks. But, these are costly as well. While SSDs serve an equivalent function as hard drives, their internal components are much different. Unlike hard drives, SSDs don't have any moving parts and thus they're called solid-state drives. Instead of storing data on magnetic platters, SSDs store data using non-volatile storage. Since SSDs haven't any moving parts, they do not need to "spin up". It ranges from 150GB to a few more TB.
- **SD Card:** It is known as a Secure Digital Card. It is generally used with electronic devices like phones, digital cameras, etc. to store larger data. It is portable and the size of the SD card is also small so that it can easily fit into electronic devices. It is available in different sizes like 2GB, 4GB, 8GB, etc.
- **Memory Card:** It is generally used in digital cameras, printers, game consoles, etc. It is also used to store large amounts of data and is available in different sizes. To run a memory card on a computer you require a separate memory card reader.
- **Multimedia Card:** It is also known as MMC. It is an integrated circuit that is generally used in-car radios, digital cameras, etc. It is an external device to store data/information.

4. Optical Storage Devices

Optical Storage Devices is also secondary storage device. It is a removable storage device. Following are some optical storage devices:

- **CD:** It is known as Compact Disc. It contains tracks and sectors on its surface to store data. It is made up of polycarbonate plastic and is circular in shape. CD can store data up to 700MB. It is of two types:
 - **CD-R:** It stands for Compact Disc read-only. In this type of CD, once the data is written can not be erased. It is read-only.
 - **CD-RW:** It stands for Compact Disc Read Write. In this type of CD, you can easily write or erase data multiple times.
- **DVD:** It is known as Digital Versatile Disc. DVDs are circular flat optical discs used to store data. It comes in two different sizes one is 4.7GB single-layer discs and another one is 8.5GB double-layer discs. DVDs look like CDs but the storage capacity of DVDs is more than as compared to CDs. It is of two types:

- **DVD-R:** It stands for Digital Versatile Disc read-only. In this type of DVD, once the data is written can not be erased. It is read-only. It is generally used to write movies, etc.
- **DVD-RW:** It stands for Digital Versatile Disc Read Write. In this type of DVD, you can easily write or erase data multiple times.
- **Blu-ray Disc:** It is just like CD and DVD but the storage capacity of blu ray is up to 25GB. To run a Blu-ray disc you need a separate Blu-ray reader. This Blu-ray technology is used to read a disc from a blue-violet laser due to which the information is stored in greater density with a longer wavelength.

5. Cloud and Virtual Storage

Nowadays, secondary memory has been upgraded to virtual or cloud storage devices. We can store our files and other stuff in the cloud and the data is stored for as long as we pay for the cloud storage. There are many companies that provide cloud services largely Google, Amazon, Microsoft, etc. We can pay the rent for the amount of space we need and we get multiple benefits out of it. Though it is actually being stored in a physical device located in the data centers of the service provider, the user doesn't interact with the physical device and its maintenance. For example, Amazon Web Services offers AWS S3 as a type of storage where users can store data virtually instead of being stored in physical hard drive devices. These sorts of innovations represent the frontier of where storage media goes.

Characteristics of Computer Storage Devices

- Data stored in the Memory can be changed or replaced in case of a requirement, because of the mobility of the storage devices.
- Storage Devices validate that saved data can be replaced or deleted as per the requirements because the storage devices are easily readable, writeable, and rewritable.
- Storage Devices are easy and convenient to access because they do not require much skill set to handle these resources.
- The storage capacity of these devices is an extra advantage to the system.
- Storage Devices have better performance and data can be easily transferred from one device to another.

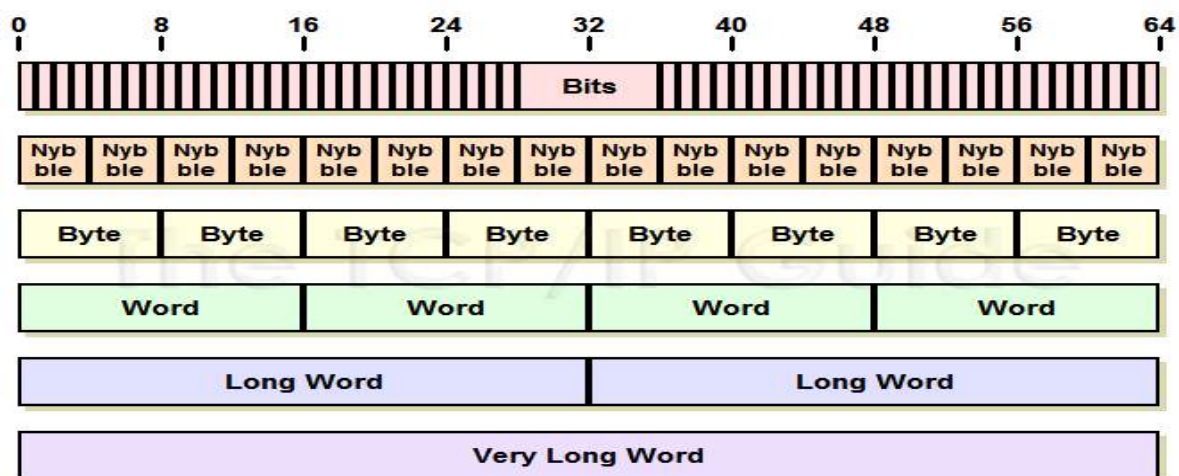
Binary Information and Representation: Bits, Bytes, Nibbles, Octets and Characters

- The fundamental building block of computer information is the *bit* (a contraction of *binary digit*). Every bit can be either 0 or 1. Making the value of a bit 1 is commonly called *setting* the bit; changing it to 0 is *resetting* or *clearing* it.
- Of course, bits represent only a very small amount of information: a single fact or value. We must make collections of these bits so we can use them to store large amounts of information and more complex data types. The most common grouping is to take eight bits and reference them as a single unit. A collection of eight bits is technically called an *octet*, but is more commonly called a *byte*. (More on that in a moment.)
- “Byte” is a jocular play on the term “bit”. Over time, other sizes of “bit collections” have also been defined. Some geek comedian decided that if eight bits made a “byte”, then four bits must be a “nybble” (or “nibble”).

Table 1: Binary Information Group Representations and Terms

Number of Bits	Common Representation Terms
1	Bit / Digit / Flag
4	Nybble / Nibble
8	Byte / Octet / Character
16	Double Byte / Word
32	Double Word / Long Word
64	Very Long Word

- A few of these terms are worth special mention. Bit and byte we have already discussed, of course. A bit is also sometimes called a *flag*; this term is most often heard when a bit is used by itself to represent a particular information state. For example, a computer might use a “changed flag” to represent whether a particular file has been modified; this is an analogy to a flag either being raised or lowered to indicate a condition. These “flags” are often seen in networking message formats.
- The term *character* is also used to express a set of eight bits. This use comes from the fact that computers often store alphanumeric characters, such as letters and numbers, one to a byte. The 16-bit *word* is fairly often used, but not nearly as much as “byte”. The larger collections of bits, such as double word and so on, are not often encountered in every-day parlance; they are used to represent chunks of data in technical fields such as hardware design or programming.



Binary Information Representations and Terms

Number System

The technique to represent and work with numbers is called Number system. Decimal number system is the most common number system. Other popular number systems include binary number system, octal number system, hexadecimal number system, etc.

Decimal Number System

Decimal number system is a base 10 number system having 10 digits from 0 to 9. This means that any numerical quantity can be represented using these 10 digits. Decimal number system is also a positional value system. This means that the value of digits will depend on its position. Let us take an example to understand this.

Say we have three numbers – 734, 971 and 207. The value of 7 in all three numbers is different–

- In 734, value of 7 is 7 hundreds or 700 or 7×100 or 7×10^2
- In 971, value of 7 is 7 tens or 70 or 7×10 or 7×10^1
- In 207, value of 7 is 7 units or 7 or 7×1 or 7×10^0

The weightage of each position can be represented as follows –

10^5	10^4	10^3	10^2	10^1	10^0
--------	--------	--------	--------	--------	--------

In digital systems, instructions are given through electric signals; variation is done by varying the voltage of the signal. Having 10 different voltages to implement decimal number system in digital equipment is difficult. So, many number systems that are easier to implement digitally have been developed. Let's look at them in detail.

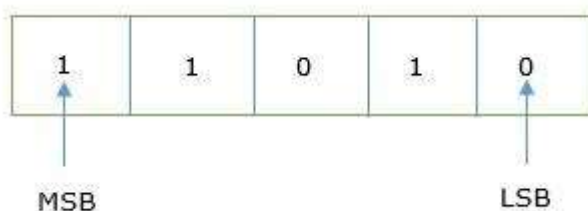
Binary Number System

The easiest way to vary instructions through electric signals is two-state system – on and off. On is represented as 1 and off as 0, though 0 is not actually no signal but signal at a lower voltage. The number system having just these two digits – 0 and 1 – is called binary number system.

Each binary digit is also called a bit. Binary number system is also positional value system, where each digit has a value expressed in powers of 2, as displayed here.

2^5	2^4	2^3	2^2	2^1	2^0
-------	-------	-------	-------	-------	-------

In any binary number, the rightmost digit is called least significant bit (LSB) and leftmost digit is called most significant bit (MSB).



And decimal equivalent of this number is sum of product of each digit with its positional value.

$$11010_2 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 16 + 8 + 0 + 2 + 0$$

$$= 26_{10}$$

Computer memory is measured in terms of how many bits it can store. Here is a chart for memory capacity conversion.

- 1 byte (B) = 8 bits
- 1 Kilobytes (KB) = 1024 bytes
- 1 Megabyte (MB) = 1024 KB
- 1 Gigabyte (GB) = 1024 MB
- 1 Terabyte (TB) = 1024 GB
- 1 Exabyte (EB) = 1024 PB
- 1 Zettabyte = 1024 EB
- 1 Yottabyte (YB) = 1024 ZB

Octal Number System

Octal number system has eight digits – 0, 1, 2, 3, 4, 5, 6 and 7. Octal number system is also a positional value system with where each digit has its value expressed in powers of 8, as shown here –

8^5	8^4	8^3	8^2	8^1	8^0
-------	-------	-------	-------	-------	-------

Decimal equivalent of any octal number is sum of product of each digit with its positional value.

$$726_8 = 7 \times 8^2 + 2 \times 8^1 + 6 \times 8^0$$

$$= 448 + 16 + 6$$

$$= 470_{10}$$

Hexadecimal Number System

Octal number system has 16 symbols – 0 to 9 and A to F where A is equal to 10, B is equal to 11 and so on till F. Hexadecimal number system is also a positional value system with where each digit has its value expressed in powers of 16, as shown here –

16^5	16^4	16^3	16^2	16^1	16^0
--------	--------	--------	--------	--------	--------

Decimal equivalent of any hexadecimal number is sum of product of each digit with its positional value.

$$27FB_{16} = 2 \times 16^3 + 7 \times 16^2 + 15 \times 16^1 + 10 \times 16^0$$

$$= 8192 + 1792 + 240 + 10$$

$$= 10234_{10}$$

ASCII

Besides numerical data, computer must be able to handle alphabets, punctuation marks, mathematical operators, special symbols, etc. that form the complete character set of English language. The complete set of characters or symbols are called alphanumeric codes. The complete alphanumeric code typically includes -

- 26 upper case letters
- 26 lower case letters
- 10 digits
- 7 punctuation marks
- 20 to 40 special characters

Now a computer understands only numeric values, whatever the number system used. So all characters must have a numeric equivalent called the alphanumeric code. The most widely used alphanumeric code is American Standard Code for Information Interchange (ASCII). ASCII is a 7-bit code that has 128 (2⁷) possible codes.

ISCII

ISCII stands for Indian Script Code for Information Interchange. ISCII was developed to support Indian languages on computer. Language supported by ISCII include Devanagari, Tamil, Bangla, Gujarati, Gurmukhi, Tamil, Telugu, etc. ISCII is mostly used by government departments and before it could catch on, a new universal encoding standard called Unicode was introduced.

Unicode

Unicode is an international coding system designed to be used with different language scripts. Each character or symbol is assigned a unique numeric value, largely within the framework of ASCII. Earlier, each script had its own encoding system, which could conflict with each other.

Boolean Operators

What Is Boolean Logic?

Boolean logic refers to the form of algebra where the variables have only 2 unique values i.e. TRUE or FALSE or ON and OFF. These values are often used as 1 or 0 in binary language or High and low logic respectively.

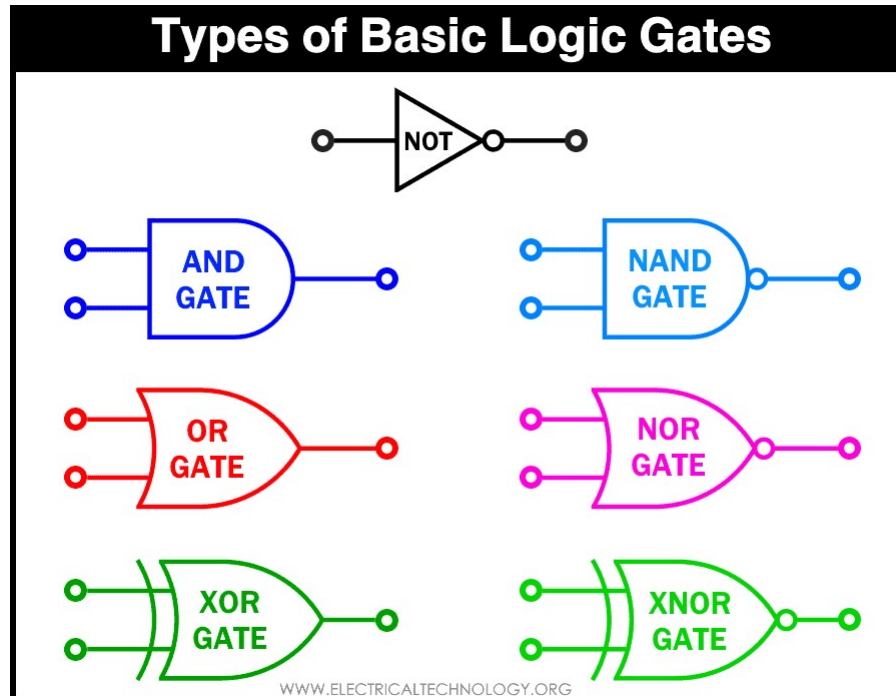
1	0
True	False
High	Low
ON	OFF

Boolean Function

A Boolean function is a logical operation of one or more than one variables whose resultant is a single binary bit. It can only be either TRUE or FALSE. Boolean functions are based on Boolean logic.

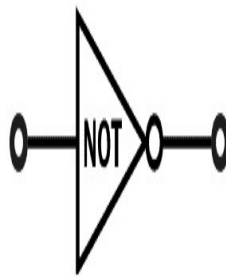
What is a Digital Logic Gate?

A digital logic gate is an electronic component which implements a Boolean function. these logic gates may have two or more than two binary inputs and provides a single binary output. Some of these basic logic gates are given below:



Types of Basic Logic Gates

NOT Gate



NOT gate is a basic digital logic gate. This logic gate inverts its input logic i.e LOW into HIGH and HIGH into LOW. It is also known as logic Inverter & is also sometimes referred to as negation buffer. NOT gate is a single input single output gate. The logical operator for NOT is '!' and it inverts its operand's value. The Truth table for NOT gate is:

Input	Output
1	0
0	1

AND Gate



This digital logic gate implements the logical AND function, which is the Boolean product of two or more than two variables. AND operation is also known as a logical conjunction. In other words, the output of AND gate is TRUE when all of its inputs are TRUE.

AND function's operator, the logical conjunction is denoted by ' $\wedge$ ' or '·'.

AND gate has a minimum of two inputs and a single output. The truth table for AND gate is:

Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0

1	1	1
---	---	---

OR Gate:



Digital logic OR Gate implements the logical OR function. Logical OR function gives TRUE output if any of its operands are TRUE. Thus, the output of OR gate is True if any of its Input is TRUE.

OR logic function is also known as inclusive disjunction. And its operator is 'v' or '+'. OR gate has a minimum of two inputs and a single output. The truth table for OR gate is:

Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate:



NAND Gate is a digital logic gate which performs negative AND function. As its name suggests, NAND (NOT of AND) operation inverts the output of AND operation. In other words, the output of the NAND gate is Low only when all of its inputs are high.

NAND gate has a minimum of two inputs and a single output.

NAND gate's truth table is

Input 1	Input 2	Output
0	0	1
0	1	1
1	0	1

1	1	0
---	---	---

NOR Gate:



The digital NOR gate is a Negative-OR gate. The operation of NOR is negation or NOT of logical OR gate. In other words, the output of NOR gate is LOW when any of its input is HIGH and vice versa.

It has a minimum of two inputs and a single output.

The truth table of NOR gate is

Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	0

XOR GATE:



Exclusive-OR or XOR gate is a digital logic gate used as a parity checker. XOR gate provides output TRUE when the numbers of TRUE inputs are odd. For a two-input XOR gate, the output is TRUE if the inputs are different. Otherwise, the gate will produce FALSE output.

The truth table of XOR gate is following

Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	0

XNOR GATE:



XNOR gate or Exclusive NOR gate is the negative or inverse of XOR gate. Generally, XNOR gate give output TRUE if it has EVEN number of TRUE inputs.

The output of two-input XNOR gate is TRUE if the inputs are same. When the inputs are different, the 2-input XNOR gate produces FALSE output.

The following is the Truth table for XNOR gate

Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	1

Concept of Algorithm and Flowchart

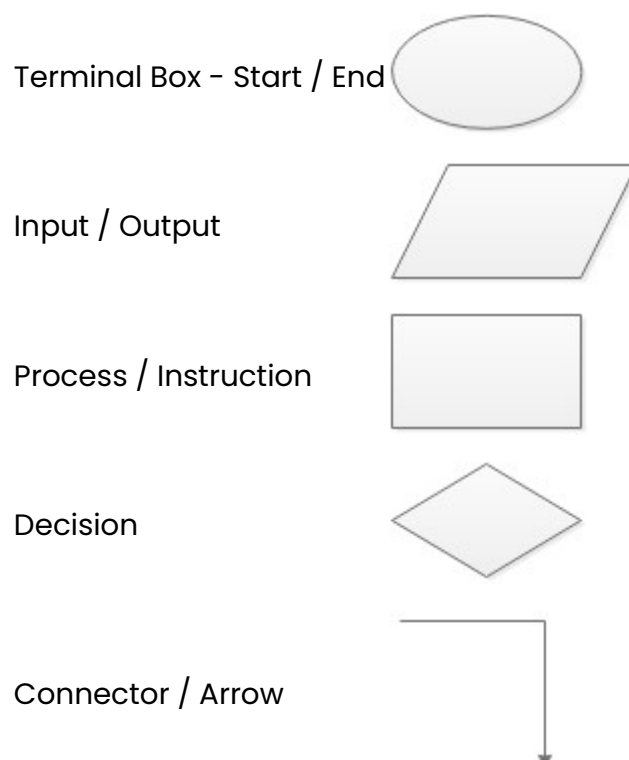
The algorithm and flowchart are two types of tools to explain the process of a program. Algorithms and flowcharts are two different tools that are helpful for creating new programs, especially in computer programming. An algorithm is a step-by-step analysis of the process, while a flowchart explains the steps of a program in a graphical way.

Writing a logical step-by-step method to solve the problem is called the algorithm. In other words, an algorithm is a procedure for solving problems. In order to solve a mathematical or computer problem, this is the first step in the process.

An algorithm includes calculations, reasoning, and data processing. Algorithms can be presented by natural languages, pseudocode, and flowcharts, etc.

A flowchart is the graphical or pictorial representation of an algorithm with the help of different symbols, shapes, and arrows to demonstrate a process or a program. With algorithms, we can easily understand a program. The main purpose of using a flowchart is to analyze different methods. Several standard symbols are applied in a flowchart:

Common Abbreviations Used in flowchart :



The symbols above represent different parts of a flowchart. The process in a flowchart can be expressed through boxes and arrows with different sizes and colors. In a flowchart, we can easily highlight certain elements and the relationships between each part.

[Difference between Algorithm and Flowchart](#)

If you compare a flowchart to a movie, then an algorithm is the story of that movie. In other words, an algorithm is the core of a flowchart. Actually, in the field of computer programming, there are many differences between algorithm and flowchart regarding various aspects, such as the accuracy, the way they display, and the way people feel about them. Below is a table illustrating the differences between them in detail.

Algorithm

- It is a procedure for solving problems.
- The process is shown in step-by-step instruction.
- It is complex and difficult to understand.
- It is convenient to debug errors.
- The solution is showcased in natural language.
- It is somewhat easier to solve complex problem.
- It costs more time to create an algorithm.

Flowchart

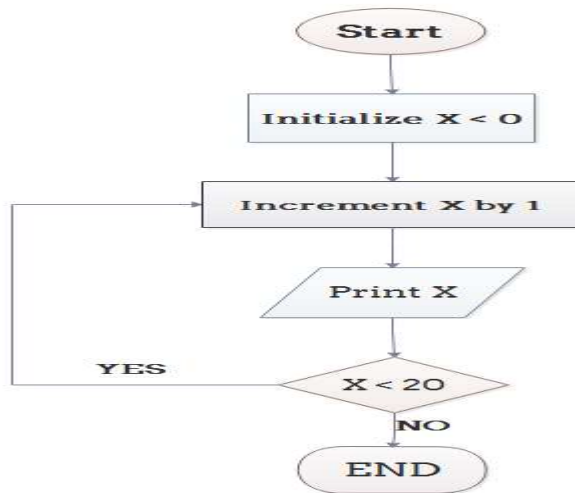
- It is a graphic representation of a process.
- The process is shown in block-by-block information diagram.
- It is intuitive and easy to understand.
- It is hard to debug errors.
- The solution is showcased in pictorial format.
- It is hard to solve complex problem.
- It costs less time to create a flowchart.

Example 1: Print 1 to 20:

Algorithm:

- Step 1: Initialize X as 0,
- Step 2: Increment X by 1,
- Step 3: Print X,
- Step 4: If X is less than 20 then go back to step 2.

Flowchart:

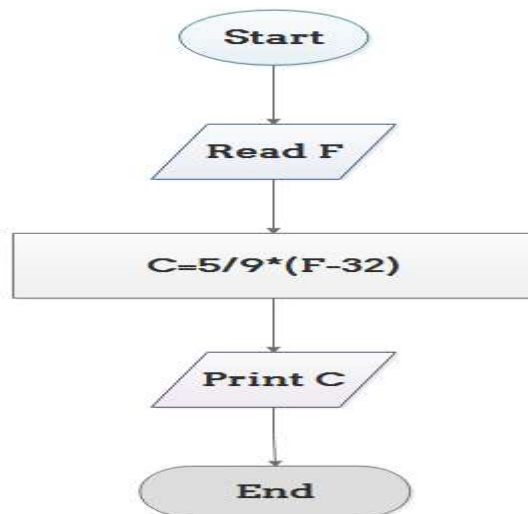


Example 2: Convert Temperature from Fahrenheit (°F) to Celsius (°C)

Algorithm:

- Step 1: Read temperature in Fahrenheit,
- Step 2: Calculate temperature with formula $C = 5/9 * (F - 32)$,
- Step 3: Print C.

Flowchart:

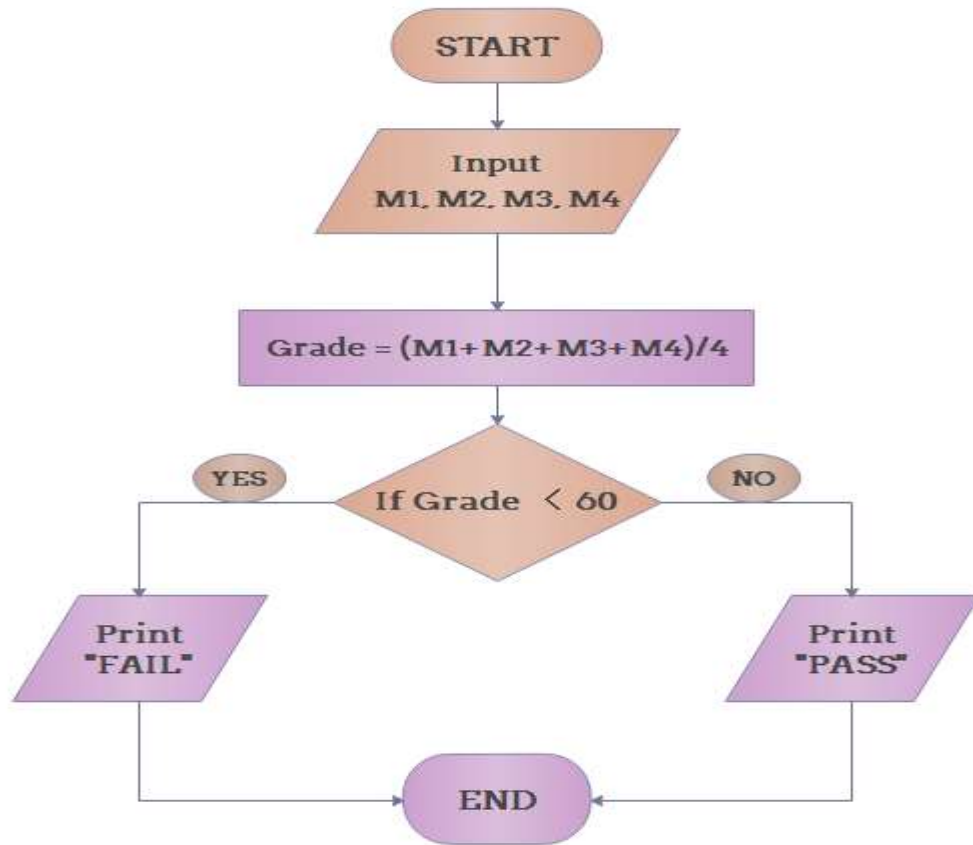


Example 3: Determine Whether A Student Passed the Exam or Not:

Algorithm:

- Step 1: Input grades of 4 courses M1, M2, M3 and M4,
- Step 2: Calculate the average grade with formula $\text{Grade} = (M1 + M2 + M3 + M4) / 4$
- Step 3: If the average grade is less than 60, print "FAIL", else print "PASS".

Flowchart:



Basics of Computer Programming

Programming Language

As we know, to communicate with a person, we need a specific language, similarly to communicate with computers, programmers also need a language is called Programming language.

Before learning the programming language, let's understand what is language?

What is Language?

Language is a mode of communication that is used to share ideas, opinions with each other. For example, if we want to teach someone, we need a language that is understandable by both communicators.

What is a Programming Language?

A programming language is a computer language that is used by programmers (developers) to communicate with computers. It is a set of instructions written in any specific language (C, C++, Java, Python) to perform a specific task.

A programming language is mainly used to develop desktop applications, websites, and mobile applications.

Types of programming language

1. Low-level programming language

Low-level language is machine-dependent (0s and 1s) programming language. The processor runs low-level programs directly without the need of a compiler or interpreter, so the programs written in low-level language can be run very fast.

Low-level language is further divided into two parts -

i. Machine Language

Machine language is a type of low-level programming language. It is also called as machine code or object code. Machine language is easier to read because it is normally displayed in binary or hexadecimal form (base 16) form. It does not require a translator to convert the programs because computers directly understand the machine language programs.

The advantage of machine language is that it helps the programmer to execute the programs faster than the high-level programming language.

ii. Assembly Language

Assembly language (ASM) is also a type of low-level programming language that is designed for specific processors. It represents the set of instructions in a symbolic and human-understandable form. It uses an assembler to convert the assembly language to machine language.

The advantage of assembly language is that it requires less memory and less execution time to execute a program.

2. High-level programming language

High-level programming language (HLL) is designed for developing user-friendly software programs and websites. This programming language requires a compiler

or interpreter to translate the program into machine language (execute the program).

The main advantage of a high-level language is that it is easy to read, write, and maintain.

High-level programming language includes Python, Java, JavaScript, PHP, C#, C++, Objective C, Cobol, Perl, Pascal, LISP, FORTRAN, and Swift programming language.

A high-level language is further divided into three parts -

i. Procedural Oriented programming language

Procedural Oriented Programming (POP) language is derived from structured programming and based upon the procedure call concept. It divides a program into small procedures called routines or functions.

Procedural Oriented programming language is used by a software programmer to create a program that can be accomplished by using a programming editor like IDE, Adobe Dreamweaver, or Microsoft Visual Studio.

The advantage of POP language is that it helps programmers to easily track the program flow and code can be reused in different parts of the program.

Example: C, FORTRAN, Basic, Pascal, etc.

ii. Object-Oriented Programming language

Object-Oriented Programming (OOP) language is based upon the objects. In this programming language, programs are divided into small parts called objects. It is used to implement real-world entities like inheritance, polymorphism, abstraction, etc in the program to makes the program reusable, efficient, and easy-to-use.

The main advantage of object-oriented programming is that OOP is faster and easier to execute, maintain, modify, as well as debug.

Example: C++, Java, Python, C#, etc.

iii. Natural language

Natural language is a part of human languages such as English, Russian, German, and Japanese. It is used by machines to understand, manipulate, and interpret human's language. It is used by developers to perform tasks such as translation, automatic summarization, Named Entity Recognition (NER), relationship extraction, and topic segmentation.

The main advantage of natural language is that it helps users to ask questions in any subject and directly respond within seconds.

3. Middle-level programming language

Middle-level programming language lies between the low-level programming language and high-level programming language. It is also known as the intermediate programming language and pseudo-language.

A middle-level programming language's advantages are that it supports the features of high-level programming, it is a user-friendly language, and closely related to machine language and human language.

Example: C, C++, language

Most commonly used Programming Language

As we all know, the programming language makes our life simpler. Currently, all sectors (like education, hospitals, banks, automobiles, and more) completely depend upon the programming language.

There are dozens of programming languages used by the industries. Some most widely used programming languages are given below -

1. Python

Python is one of the most widely used user-friendly programming languages. It is an open-source and easy to learn programming language developed in the 1990s. It is mostly used in Machine Learning, Artificial Intelligence, Big Data, GUI based desktop applications, and Robotics.

Advantages

- Python is easy to read, easy to understand, and easy to write.
- It integrates with other programming languages like C, C++, and Java.
- Python executes code line-by-line, so it is easy for the programmer to find the error that occurred in the code.
- Python is platform-independent means you can write code once and run it anywhere.

Disadvantages

- Python is not suitable for developing mobile applications and games.
- Python works with the interpreter. That's why it is slower than other programming languages like C and C++.

2. Java

Java is a simple, secure, platform-independent, reliable, architecture-neutral high-level programming language developed by Sun Microsystems in 1995. Now, Java is owned by Oracle. It is mainly used to develop bank, retail, information technology, android, big data, research community, web, and desktop applications.

Advantages

- Java is easy to write, compile, learn, and debug as compared to other programming languages.
- It provides an ability to run the same program on different platforms.
- It is a highly secured programming language because in java, there is no concept of explicit pointers.
- It is capable of performing multiple tasks at the same time.

Disadvantages

- Java consumes more memory and slower than other programming languages like C or C++.
- It does not provide a backup facility.

3. C

C is a popular, simple, and flexible general-purpose computer programming language. Dennis M Ritchie develops it in 1972 at AT&T. It is a combination of both low-level programming language as well as a high-level programming language. It is used to design applications like Text Editors, Compilers, Network devices, and many more.

Advantages

- C language is easy to learn.
- It is fast, efficient, portable, easy to extend, powerful, and flexible programming language.
- It is used to perform complex calculations and operations such as MATLAB.
- It provides dynamic memory allocation to allocate memory at the run time.

Disadvantages

- In the C programming language, it is very difficult to find the errors.
- C does not support the concepts of constructors, destructors, abstraction, polymorphism, encapsulation, and namespace like OOPs.

4. C++

C++ is one of the thousands of programming languages that we use to develop software. C++ programming language is developed by Bjarne Stroustrup in 1980. It is similar to the C programming language but also includes some additional features such as exception handling, object-oriented programming, type checking, etc.

Advantages

- C++ is a simple and portable structured programming language.
- It supports OOPs features such as Abstraction, Inheritance, Encapsulation.
- It provides high-level abstraction and useful for a low-level programming language, and more efficient for general-purpose.
- C++ is more compatible with the C language.

Disadvantages

- C++ programming language is not secured as compared to other programming languages like Java or Python.
- C++ can not support garbage collection.
- It is difficult to debug large as well as complex web applications.

5. C#

C# (pronounced as C sharp) is a modern, general-purpose, and object-oriented programming language used with XML based Web services on the .NET platform. It is mainly designed to improve productivity in web applications. It is easier to learn for those users who have sufficient knowledge of common programming languages like C, C++, or Java.

Advantages

- C# is a modern, type-safe, easy, fast, and open-source programming language that is easily integrated with Windows.
- The maintenance of C# (C sharp) is lower than the C++ programming language.
- C# is a pure object-oriented programming language.
- C# includes a strong memory backup facility. That's why it avoids the problem of memory leakage.

Disadvantages

- C# is less flexible because it is completely based on Microsoft .Net framework.
- In C#, it is difficult to write, understand, debug, and maintain multithreaded applications.

6. JavaScript

Javascript is a type of scripting language that is used on both client-side as well as a server-side. It is developed in the 1990s for the Netscape Navigator web browser. It allows programmers to implement complex features to make web pages alive. It helps programmers to create dynamic websites, servers, mobile applications, animated graphics, games, and more.

Advantage

- JavaScript helps us to add behavior and interactivity on the web page.
- It can be used to decrease the loading time from the server.
- It has the ability to create attractive, dynamic websites, and rich interfaces.
- JavaScript is a simple, versatile, and lightweight programming language.
- JavaScript and its syntax are easy to understand.

Disadvantage

- JavaScript is completely based on the browser.
- It does not support multiple inheritance.
- It is less secure compared to other programming languages.

7. R

Currently, R programming is one of the popular programming languages that is used in data analytics, scientific research, machine learning algorithms, and statistical computing. It is developed in 1993 by Ross Ihaka and Robert Gentleman. It helps marketers and data scientists to easily analyze, present, and visualize data.

Advantages

- R programming provides extensive support for Data Wrangling.
- It provides an easy-to-use interface.
- It runs on any platform like Windows, Linux, and Mac.
- It is an open-source and platform-independent programming language.

Disadvantages

- R programming does not support 3D graphics.
- It is slower than other programming languages.

8. PHP

PHP stands for Hypertext Preprocessor. It is an open-source, powerful server-side scripting language mainly used to create static as well as dynamic websites. It is developed by Rasmus Laird in 1994. Inside the php, we can also write HTML, CSS, and JavaScript code. To save php file, file extension .php is used.

Advantages

- PHP is a more secure and easy-to-use programming language.
- It supports powerful online libraries.
- It can be run on a variety of operating systems such as Windows, Linux, and Mac.
- It provides excellent compatibility with cloud services.

Disadvantages

- PHP is not capable of handling a large number of applications and not suitable for large applications.
- It is quite difficult to maintain.

9. Go

Go or Golang is an open-source programming language. It is used to build simple, reliable, and efficient software. It is developed by Robert Griesemer, Rob Pike, and Ken Thompson in 2007.

Advantages

- Go language is easy-to-learn and use.
- It comes with the in-built testing tools.
- Go is a fast programming language.

Disadvantages

- Go language does not support generics.
- It does not support error handling.
- It supports a lack of frameworks.

10. Ruby

Ruby is an open-source, general-purpose, and pure object-oriented programming language released in 1993. It is used in front-end and back-end web development. It is mainly designed to write CGI (Common Gateway Interface) scripts.

Advantages

- Ruby supports various GUI (Graphical User Interface) tools like GTK and OpenGL.
- It is used to develop both internet as well as intranet applications.
- The code written in Ruby is small and contains less number of lines.

Disadvantages

- Ruby is slower than other programming languages.
- It is very difficult for programmers to debug the code written in Ruby.

Difference Between Procedural and Object Oriented Programming

Parameter	Procedural Programming	Object Oriented Programming
Definition	This programming language makes use of a step by step approach for breaking down a task into a collection of routines (or subroutines) and variables by following a sequence of instructions. It carries out each step systematically in order so that a computer easily gets to understand what to do.	This programming language uses objects and classes for creating models based on the real-world environment. This model makes it very easy for a user to modify as well as maintain the existing code while new objects get created by inheriting the characteristics of the present ones.
Security	Procedural Programming does not offer any method of hiding data. Thus, it is less secure when	Hiding data is possible with Object Oriented Programming due to the

	compared to Object Oriented Programming.	abstraction. Thus, it is more secure than the Procedural Programming.
Method	The main program gets divided into minute parts on the basis of the functions. It then treats them as separate programs for smaller programs individually.	It involves the concept of classes and objects. Hence, it divides the program into minute chunks known as objects. These are actually instances of classes.
Division of Program	Procedural Programming divides the program into small programs and refers to them as functions.	Object Oriented Programming divides the program into small parts and refers to them as objects.
Movement of Data	Available data is capable of moving freely within the system from one function to another.	The objects are capable of moving and communicating with each other through the member functions.
Approach	The Procedural Programming follows a Top-Down approach.	The Object Oriented Programming follows a Bottom-Up approach.
Importance	This programming model does not give importance to data. It prioritizes the functions along with the sequence of actions that needs to follow.	This programming model gives importance to the data rather than functions or procedures. It is because it works on the basis of the real world.
Orientation	It is Structure/Procedure oriented.	It is Object Oriented.

Basis	The main focus in Procedural Programming is on how to do the task, meaning, on the structure or procedure of the program.	The main focus in Object Oriented Programming is on data security. Hence, it only permits objects to access the class entities.
Type of Division	It divides any large program into small units called functions.	It divides the entire program into small units called objects.
Inheritance	It does not provide any inheritance.	It achieves inheritance in three modes- protected, private, and public.
Virtual Classes	There is no concept of virtual classes.	The concept of virtual functions appears at the time of inheritance.
Overloading	The case of overloading isn't possible in the case of Procedural Programming.	Overloading is possible in the form of operator overloading and function overloading in the case of Object Oriented Programming.
Reusability of Code	No feature of reusing codes is present in Procedural Programming.	Object Oriented Programming offers the feature to reuse any existing codes in it by utilizing a feature known as inheritance.
Most Important Attribute	It prioritizes function over data.	It prioritizes data over function.

Modes of Access	The Procedural Programming offers no specific accessing mode for accessing functions or attributes in a program.	The Object Oriented Programming offers three accessing modes- protected, private, and public. These, then, serve as a share to access functions of attributes.
Size of Problems	It is not very suitable for solving any big or complex problems.	It is suitable for solving any big or complex problems.
Addition of New Function and Data	It is not very easy to add new functions and data in the Procedural Programming.	It is very easy to add new functions and data in the Object Oriented Programming.
Access to Data	In the Procedural Programming, most of the functions use global data for sharing. They can access freely from one function to another in any given system.	In the Object Oriented Programming, the present data cannot easily move easily from one function to another. One can keep it private or even public. Thus, a user can control the data access.
Data Sharing	It shares the global data among the functions present in the program.	It shares data among the objects through its member functions.
Data Hiding	No proper way is available for hiding the data. Thus, the data remains insecure.	It can hide data in three modes- protected, private, and public. It increases the overall data security.

Basis of World	The Procedural Programming follows an unreal world.	The Object Oriented programming follows the real world.
Friend Classes or Friend Functions	It doesn't involve any concept of friend function.	Any class or function is capable of becoming a friend of any other class that contains the keyword "friend." Note – The keyword "friend" only works for C++.
Examples	Some common examples of Procedural Programming are C, Fortran, VB, and Pascal.	The examples of Object Oriented Programming languages are Java, C++, VB.NET, Python, and C#.NET.

Compiler and Interpreter

The Compiler and Interpreter, both have similar works to perform. Interpreters and Compilers convert the Source Code (HLL) to Machine Code (understandable by Computer). In general, computer programs exist in High-Level Language that a human being can easily understand. But computers cannot understand the same high-level language, so for computers, we have to convert them into machine language and make them readable for computers. In this article, we are going to see the differences between them.

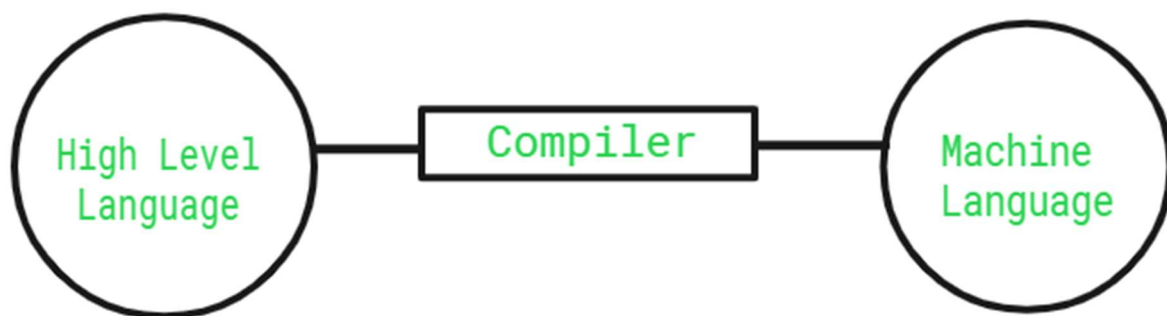
Compiler

The Compiler is a translator which takes input i.e., High-Level Language, and produces an output of low-level language i.e. machine or assembly language. The work of a Compiler is to transform the codes written in the programming language into machine code (format of 0s and 1s) so that computers can understand.

- A compiler is more intelligent than an assembler it checks all kinds of limits, ranges, errors, etc.
- But its program run time is more and occupies a larger part of memory. It has a slow speed because a compiler goes through the entire program and then translates the entire program into machine codes.

Role of a Compiler

For Converting the code written in a high-level language into machine-level language so that computers can easily understand, we use a compiler. Converts basically convert high-level language to intermediate assembly language by a compiler and then assembled into machine code by an assembler.



Advantages of Compiler

- Compiled code runs faster in comparison to Interpreted code.
- Compilers help in improving the security of Applications.
- As Compilers give Debugging tools, which help in fixing errors easily.

Disadvantages of Compiler

- The compiler can catch only syntax errors and some semantic errors.
- Compilation can take more time in the case of bulky code.

Interpreter

An Interpreter is a program that translates a programming language into a comprehensible language. The interpreter converts high-level language to an intermediate language. It contains pre-compiled code, source code, etc.

- It translates only one statement of the program at a time.
- Interpreters, more often than not are smaller than compilers.

Role of an Interpreter

The simple role of an interpreter is to translate the material into a target language. An Interpreter works line by line on a code. It also converts high level language to machine language.



Advantages of Interpreter

- Programs written in an Interpreted language are easier to debug.
- Interpreters allow the management of memory automatically, which reduces memory error risks.
- Interpreted Language is more flexible than a Compiled language.

Disadvantages of Interpreter

- The interpreter can run only the corresponding Interpreted program.
- Interpreted code runs slower in comparison to Compiled code.

Difference Between Compiler and Interpreter

Compiler	Interpreter
Steps of Programming: • Program Creation. • Analysis of language by the compiler and	Steps of Programming: • Program Creation. • Linking of files or generation of Machine Code is not required by Interpreter.

Compiler	Interpreter
throws errors in case of any incorrect statement. • In case of no error, the Compiler converts the source code to Machine Code. • Linking of various code files into a runnable program. • Finally runs a Program.	• Execution of source statements one by one.
The compiler saves the Machine Language in form of Machine Code on disks.	The Interpreter does not save the Machine Language.
Compiled codes run faster than Interpreter.	Interpreted codes run slower than Compiler.
Linking-Loading Model is the basic working model of the Compiler.	The Interpretation Model is the basic working model of the Interpreter.
The compiler generates an output in the form of (.exe).	The interpreter does not generate any output.
Any change in the source program after the compilation requires recompiling the entire code.	Any change in the source program during the translation does not require retranslation of the entire code.

Compiler	Interpreter
Errors are displayed in Compiler after Compiling together at the current time.	Errors are displayed in every single line.
The compiler can see code upfront which helps in running the code faster because of performing Optimization.	The Interpreter works by line working of Code, that's why Optimization is a little slower compared to Compilers.
It does not require source code for later execution.	It requires source code for later execution.
Execution of the program takes place only after the whole program is compiled.	Execution of the program happens after every line is checked or evaluated.
Compilers more often take a large amount of time for analyzing the source code.	In comparison, Interpreters take less time for analyzing the source code.
CPU utilization is more in the case of a Compiler.	CPU utilization is less in the case of a Interpreter.
The use of Compilers mostly happens in Production Environment.	The use of Interpreters is mostly in Programming and Development Environments.
Object code is permanently saved for future use.	No object code is saved for future use.

Compiler	Interpreter
C, C++, C# etc are programming languages that are compiler-based.	Python , Ruby, Perl, SNOBOL, MATLAB, etc are programming languages that are interpreter-based.